

Rural residents' choice of the payment grades of basic old-age insurance for urban and rural residents

Diploma Thesis

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Hereby I would like to thank my thesis supervisor Ing. Ladislava Issever Grochová, Ph.D. for her expert guidance throughout the whole writing process as well as the valuable advice that was so kindly provided.

Statutory declaration

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Abstract

Zhao J., Rural residents' choice of the payment grades of basic old-age insurance for urban and rural residents. Diploma thesis. Brno: Mendel University, 2022.

This study uses a multi-logit model to analyze the factors affecting rural residents' choice of insurance contribution brackets based on data from the 2019 Tengzhou city field survey. It was found that gender, age, marital status, education, health, number of children, annual per capita household income, total household assets, understanding of pension subsidies, understanding of the procedure for changing contribution brackets, satisfaction with the processing of insurance, and satisfaction with the basic pension payment influenced the choice of contribution brackets. Based on this, the thesis proposes corresponding policy recommendations to promote the process of equalization of urban and rural old-age insurance services.

Keywords

Payment grades, urban and rural residents old-age insurance, rural residents.

Abstrakt

Zhao J., Výběr platebních stupňů základního důchodového pojištění obyvatel měst a venkova obyvateli venkova. Diploma thesis. Brno: Mendelova univerzita v Brně, 2022.

Tato studie pracuje s multilogitovým modelem k analýze faktorů ovlivňujících volbu pojistných sazeb obyvatel z venkova na základě údajů z terénního šetření provedeného roku 2019 ve městě Teng-čou. Prokázalo se, že vliv na volbu příspěvkových pásem mají pohlaví, věk, rodinný stav, vzdělání, zdravotní stav, počet dětí, roční příjem domácnosti na osobu, celkový majetek domácnosti, porozumění dotacím na důchod, porozumění postupu při změně příspěvkových pásem, spokojenost s vyřízením pojištění a spokojenost s výplatou základního důchodu. Na základě toho práce navrhuje odpovídající politická doporučení na podporu procesu vyrovnávání služeb starobního pojištění ve městech a na venkově.

Klíčová slova

Platební třída, důchodové pojištění obyvatel měst a venkova, obyvatelé venkova.

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1 Introduction

1.1 Research background

In 1978, the government introduced a family planning policy in order to slow down the explosive growth of China's population. As the policy was implemented, rapid population growth was slowed and the family planning policy gradually became a basic state policy in China. The opening up and reform of China in the 1980s and the demographic window of opportunity interacted to turn the demographic window of opportunity into a demographic dividend, which contributed to China's rapid economic development.

But after 2010, China's demographic window of opportunity narrowed and the demographic dividend gradually disappeared. The long-standing family planning policy has led to a low fertility rate in China and a serious problem of an ageing population. The China Council on Ageing predicts that China's elderly population will grow by an average of around 10 million per year from 2020 to 2035. Additionally, the number of senior persons will rise to 418 million between 2020 and 2035, making up 28.7% of the national population. Furthermore, China is hesitant to make changes to its accumulation-based social pension system, and it will be challenging to absorb the social costs of modifying the system quickly. And China's pension investments are too homogeneous with low returns. All of these will raise payment risk in the future and are detrimental to the pension insurance system's ability to remain financially viable.

The income and spending graph for China's basic pension fund shows that there is a very little difference between the two, and that there are certain years when the fund's income is actually lower than its expenditure. As the problem of an ageing population becomes more acute, the expenditure of the pension fund will also expand, which will exacerbate the problem of uneven fund income and expenditure.

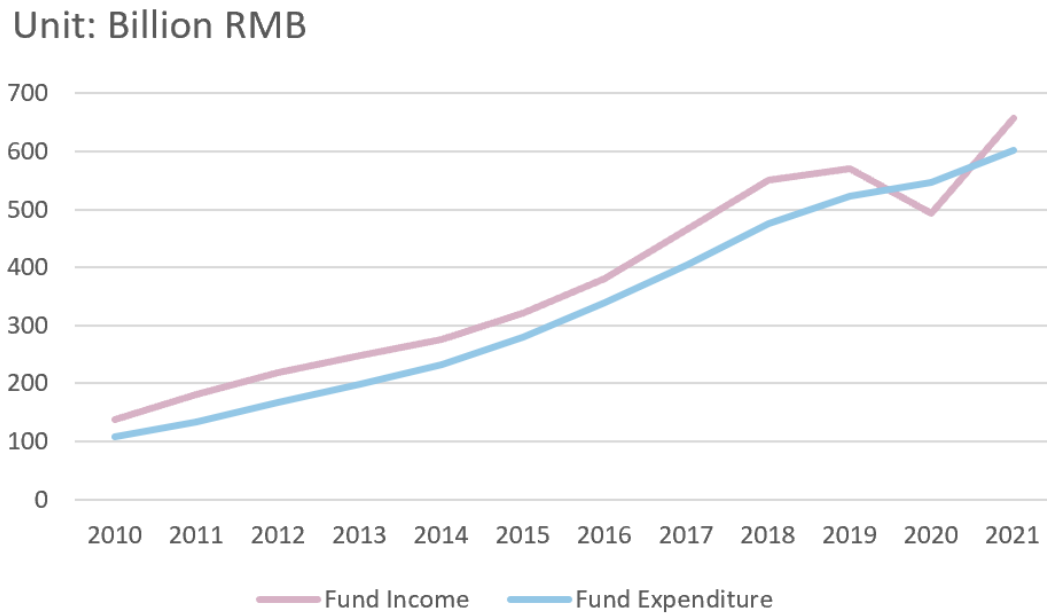


Fig.1 The incomes and expenses of China's basic old-age insurance fund

Source: Ministry of Human Resources and Social Security of the People's Republic of China, 2010-2021.

Most regions in China have only achieved provincial co-ordination of pension insurance, so funds are dispersed to individual provinces and managed by local governments. However, local governments are most concerned about the safety and liquidity of the funds, adhere to the principle of prudent investment, and the existing investment channels are narrow. As a result, local governments have mostly chosen to deposit huge amounts of pension funds in banks or use them to buy treasury bonds, earning only very low returns and failing to preserve and increase their value. The fund's rate of return up to 2014 was only about 2.5%. The National Social Security Fund Board of Trustees took over investment management duties for the insurance fund in 2015, which significantly increased management effectiveness and investment returns. But by the end of 2018, just 12% of the old-age insurance fund had been efficiently transferred to the trustee of the National Social Security Fund Trustees, leaving a sizable amount of money unutilized (Wang Z., 2021). The risk of pension devaluation has an effect on the sustainability of the old-age insurance program as well.

Finally, it is evident that China's old-age insurance program has seen significant setbacks in regard to its capacity to grow sustainably.

1.2 Tengzhou's transformation of pension insurance

Tab.1 Tengzhou urban and rural inhabitants' old-age insurance comparison

Insurance	Insurance Protection	Payment grade (person-year)	Subsidies (person-year)
Old-age insurance for urban and rural inhabitants (2010)	Inhabitants aged 16 and over who are not covered by other social old-age insurance (Not students)	Eight grades of 100, 200, 300, 400, 500, 700, 900, and 1000 RMB.	30 RMB
(2012)	Consistent with 2010	Three additional grades: 1,500, 2,000, and 3,000 RMB. But the 100 and 200 RMB have been created for people with impairments.	30 RMB
(2015)	Consistent with 2010	Twelve grades of 100, 300, 500, 600, 800, 1000, 1500, 2000, 2500, 3000, 4000, and 5000 RMB. However, the RMB 100 is only for the severely disabled and other groups with economic difficulty.	Grades of 100 and 300 RMB: 30RMB; grades of 500 RMB and above: 60 RMB
(2022)	Consistent with 2010	Consistent with 2015	Grades of 100 and 300 RMB: 30RMB; grades of 500 and 600RMB: 60 RMB; grades of 800 RMB and above: 80 RMB

Source: Tengzhou Ministry of Human Resources and Social Security, 2010,2012,2015,2022.

As shown in Tab.1, the evolution of the old-age insurance program has given rise to more flexibility in the design of the contribution brackets. Rural residents can

choose the appropriate contribution bracket according to their financial situation. From a government subsidy standpoint, the total amount of subsidies has increased and is also more detailed, reflecting a certain degree of differentiation.

1.3 The current situation of pension insurance in Tengzhou

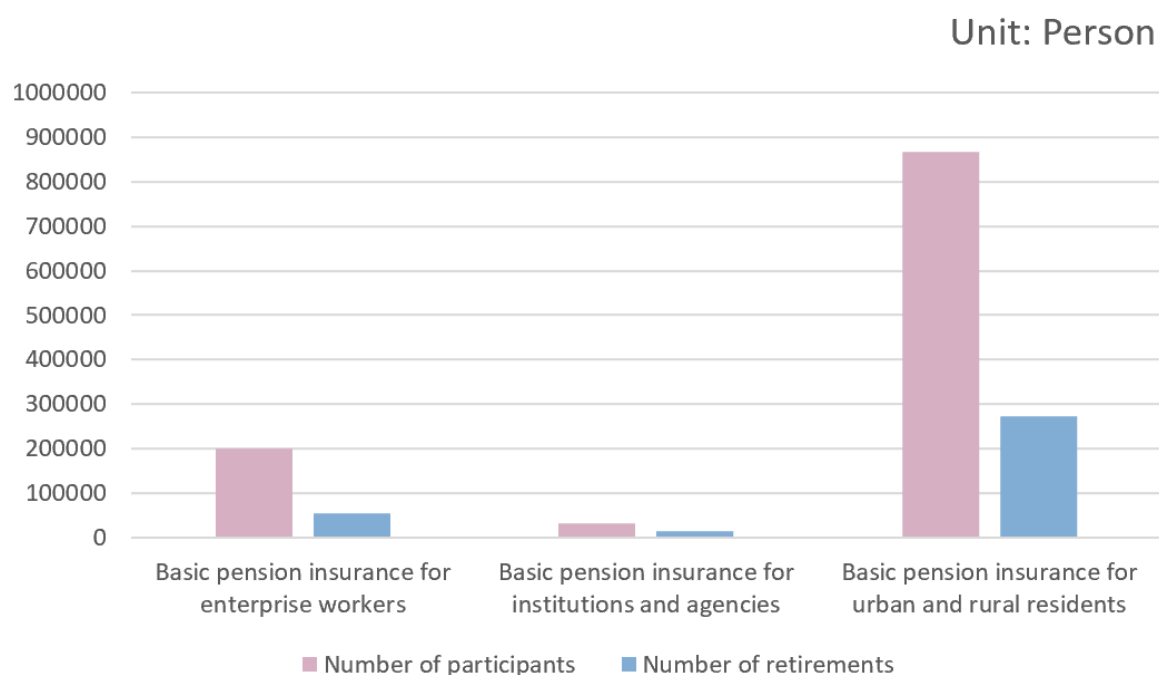


Fig.2 Number of participants and retirements in the three forms of social pension insurance by June 2022

Source: Human Resources and Social Security Bureau in Tengzhou, 2022

Figure 2 demonstrates the wide coverage of the third insurance by showing that the majority of citizens choose it and the majority of pensioner. The most significant explanation for this is because the other two types of social insurance contain restrictions on who can purchase the insurance, necessitating that they be workers of businesses or staff members of institutions. However, the total population of Tengzhou is 1.765 million, so there are still some people who are not included in the pension insurance body.

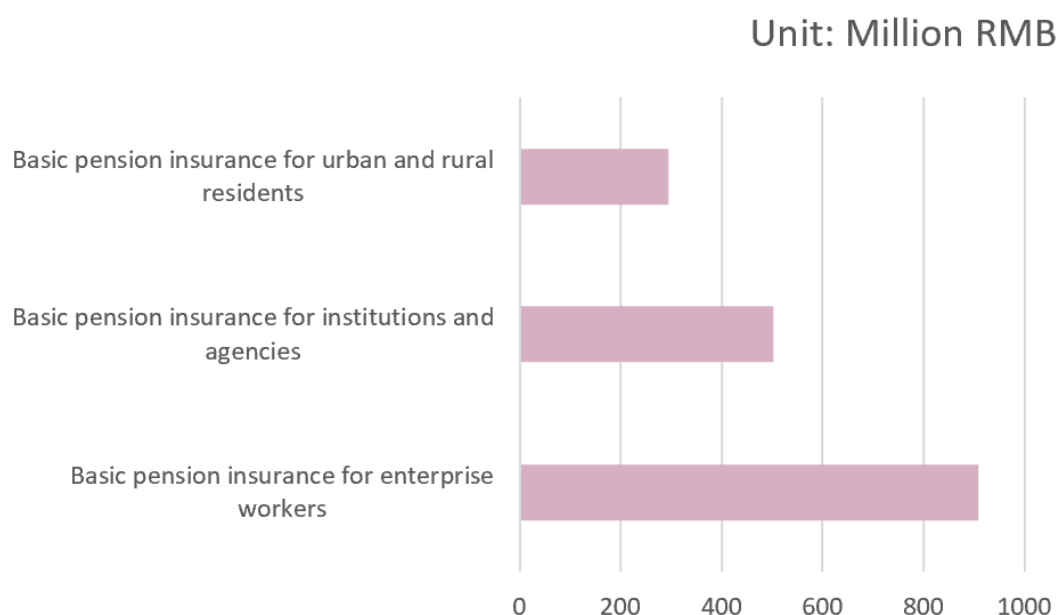


Fig.3 Accumulated expenditure in the three types of social pension insurance at the end of June 2022

Source: Human Resources and Social Security Bureau in Tengzhou, 2022

Even while the number of individuals contributing to the old-age insurance for urban and rural inhabitants is the biggest, as shown in the aforementioned graph, the total cost of this insurance is the lowest. The per capita old-age insurance benefit is only 6.55% of the old-age insurance for enterprise employees and 3.13% of the old-age insurance for public institutions, according to additional calculations. Basic old-age insurance pension is just RMB 1,074.67 per person annually for urban and rural populations. In addition, Tengzhou's rural citizens would have a monthly minimum standard of living of 700 yuan in 2022. As a result, the benefits do not fulfill the requirement for protection because they fall short of the minimum subsistence level.

1.4 Significance of study

Theoretical significance: It is discovered via gathering and scouring the literature on urban and rural inhabitants' pension insurance systems that China's system is still being researched and developed and urgently needs to be improved. The willingness to engage in this pension insurance, the circumstances surrounding participation, the accountable body, the financial subsidy standard, the implementation effect, and the incentive mechanism, among other topics, have primarily been researched by academics., but less on the choice of the rural

residents' contribution grade. The study looks at the choice of contribution band for rural inhabitants, which is helpful for creating a reliable pension insurance scheme. In addition side, it also helps to solve the challenging pension issue facing rural people, further close the urban-rural divide, eliminate the urban-rural dichotomy, and achieve social justice and long-term stability. Overall, this thesis examines the variables influencing the contribution grade of rural residents' basic pension insurance, adds to the findings of previous research on this topic, advances the theoretical underpinnings of the creation of this system, and encourages research on the pension security of rural and urban residents.

Realistic significance: The entire accumulation and operation of social pensions will be directly impacted by the choice of the contribution grade for this old-age insurance. Additionally, a sizeable portion of social old-age insurance is made up of this old-age insurance. If people frequently choose the lowest contribution grade, the government will need to raise pension financial subsidies in order to cover the residents' essential pension needs because the lowest contribution grade can rarely meet their pension demands. This will result in the government taking on an excessive amount of the duty currently held by individuals, as well as an unjustified financial load, which will have an impact on how sustainably and healthily the pension insurance system develops. This pension system is solely coordinated at the county level due to the difficult transfer of old-age insurance policies. However, the problems identified by scholars are too broad-based on the provincial perspective, and the suggestions made based on the problems are not targeted. It is challenging to replicate the research findings from other regions because the economic development levels of different regions vary.

2 Objectives

Rural residents typically select the lowest contribution category, as seen during the installation of this old-age insurance system. The viability of the system will be significantly impacted if this old-age insurance program continues to struggle with the issue of declining contributions since it won't be able to provide people with the security of the system as they age. As a result, the basic pension insurance contribution bracket and the influencing factors in Tengzhou are the research object for this thesis. The specific research areas are the below three.

- The current situation of the actual contribution level of rural residents in Tengzhou City. The data obtained from the field research was analyzed to explore whether there is a dilemma of low contribution brackets.
- To examine the variables affecting Tengzhou's rural inhabitants' decision for this old-age insurance grades. An analysis of each variable's impact on the contribution brackets was conducted using a multi-logit model.
- At the level of the individual, the government, and the system, recommendations are made for how to increase the contribution bracket for rural inhabitants in Tengzhou.

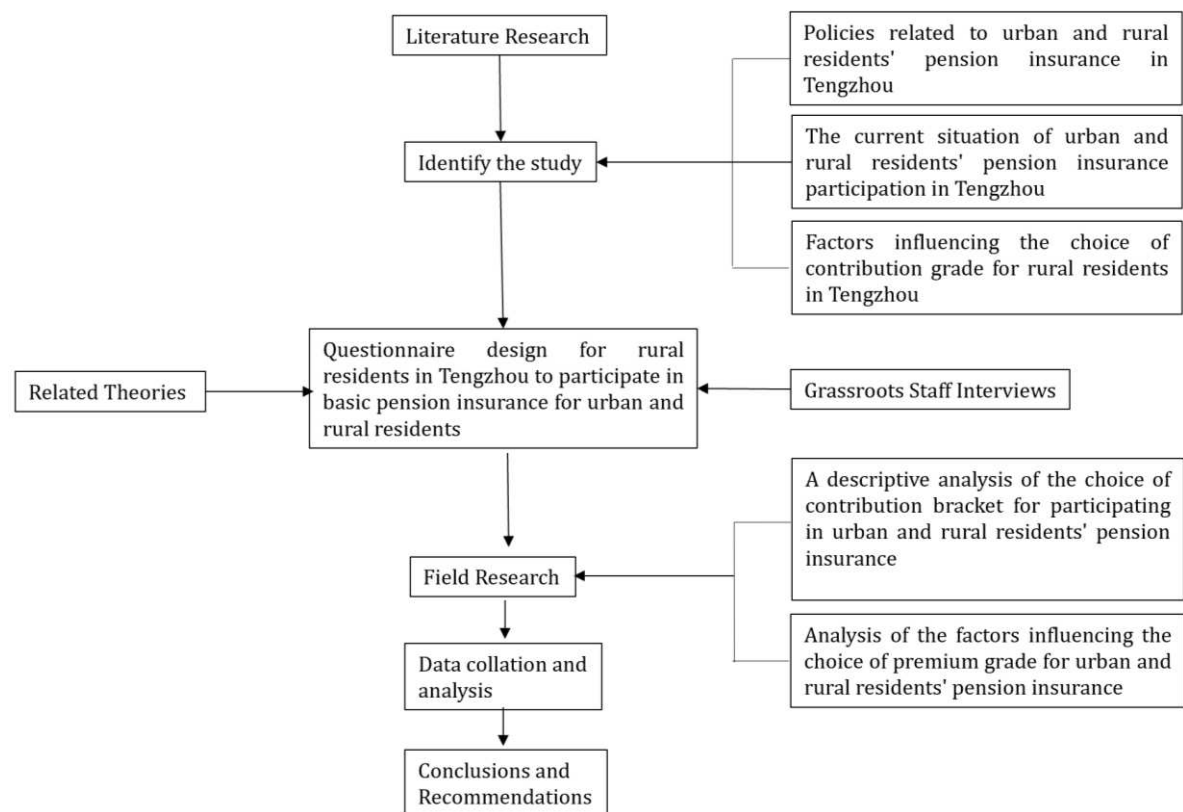


Fig.4 Technology roadmap

3 Literature review

The "Guiding Opinions on the Piloting of a New Type of Rural Social Pension Insurance System" were published by the State Council on September 4th, 2009. This opinion outlines the fundamental ideas behind the new kind of rural old-age insurance and suggests setting up a few pilot projects first. Based on the results, the areas would subsequently be expanded, and by 2020, the entire country would essentially be covered. The system specifies the standard of central and local subsidies and is a universal old-age security with the government paying the full amount of the basic old-age pension for those aged 60 and above.

The new kind of rural old-age insurance has significantly improved social security, yet there are still some serious issues. Discontinuity of coverage is more likely because of the incapacity to connect businesses with urban and rural areas, the lengthy contribution period, and the asymmetry of policy knowledge. The primary problem is that urban workers who use flexible scheduling are not protected by the social security system and are therefore unable to access social insurance welfare. According to "instructions on the pilot program of social pension insurance for urban residents" published on June 7, 2011, new rural social old-age insurance and social old-age insurance for urban citizen may be combined in areas where requirements exist. Non-working citizens who are at least 16 years old or who do not qualify for the social old-age insurance for city laborers may, if they choose, pay for this insurance at the place of their household registration. China's social pension insurance program now has nationwide coverage since that time.

Although social pension insurance covers all groups, there is a problem of fragmentation. This problem was addressed by creating a single old-age insurance scheme. A old-age insurance program for urban and rural populations has started with the deployment of this pension insurance pilot in some places. With more than three years of piloting, more valuable experience has been accumulated and its feasibility and advantages have been proven. On February 21, 2014, the State Council approved "the Opinions on the Establishment of a Unified Basic Pension Insurance System for Urban and Rural Citizens" in order to formally link the new kind of rural old-age insurance with the urban citizens old-age insurance. The implementation of this system promotes its connection to the employee pension insurance scheme and connects communities in rural and urban areas under a single insurance program. The "duality" between urban and rural structure will be

eliminated as a result, leading to improved coordination of urban and rural expansion. Pensioners from both areas will receive equal treatment, and both communities will grow together.

3.1 Government responsibility in pension insurance

Chinese academics have long been interested in China's basic pension insurance scheme for urban and rural inhabitants, but because the field of study only recently began, there is little direct literature on the government responsibility. Scholars have mostly focused on social responsibility in pension insurance, lacking in-depth analysis and research on this issue, which in summary focuses on two aspects.

3.1.1 Research on the positioning of government responsibility

Scholars in various countries generally agree that the government should assume the responsibility related to the social pension insurance system at this stage. However, it is challenging to come to an agreement on how the government should be held accountable for social security. Because this insurance is a substantial part of China's social security, a clear explanation of the government's obligations is important for the country's sustainable and healthy growth (H. Huang & Li, 2020).

Domestic and foreign scholars believe that pension insurance is divided into multi-participants together and single participants. In an active welfare society, the relationship between the individual and the government has changed, and social welfare should not be borne by the government alone, but by the government together with other institutions such as enterprises (Giddens, 2000). From the standpoint of government responsibility, it can be seen that foreign nations primarily adopt the models of full government responsibility, shared government and self responsibility, and full self responsibility, and these three different responsibility models of rural social pension system reflect the decreasing degree of government responsibility (Cao & Li, 2011). The system has been divided by researchers according on the extent of government involvement. By dividing government responsibility in terms of changes, it is found that government responsibility was absent from the founding of New China to the beginning of reform and opening up; governmental accountability started to resurface after reform and opening up in the early 21st century, but to a limited extent; and government responsibility has returned from the beginning of the 21st century to the present (X. Yang & Wang, 2019).

Chinese scholars have reached a consensus on the government's role as the main body in the pension insurance system. China's rural old-age social insurance system is at a low level and needs urgent improvement. Government should assume social responsibility and take the lead in establishing rural old-age social insurance because it is the principal provider of social security (J. Lu, 2004). Combining the government's role and realistic responsibilities in long-term pension insurance, the author believes that the main body responsible for the pension insurance system must be the government. China must adhere to the main position of the government and take fairness, efficiency, and people's satisfaction as measure to play the main role more actively in the future (Yu, 2015). The government's role is increasingly becoming more important, according to the author, and it has always taken the lead in the creation of fundamental pension insurance. In order to create a coordinated and cooperative system of duty division, the government is also stepping up cooperation between other subjects, such as the market (Lv, 2021).

For great many years, the efficiency of pension insurance has been severely hampered by the vagueness of what constitutes government obligation. Therefore, in recent years, scholars have focused on how to reasonably divide government responsibilities. The author argued that the government's responsibility can be divided into policy and financial responsibilities (Chen & Fang, 2016). Another scholar proposes that the government should assume managerial and direct fiscal responsibilities, which are relatively independent yet affect each other. Managerial responsibility involves designing the pension insurance system and ensuring its smooth operation; direct fiscal responsibility involves accurate planning to measure the amount of direct financial subsidies for pension insurance (Hong, 2018). In addition to the governmental responsibilities mentioned above, the government should also bear legislative, system design, fiscal, supervision, and management responsibilities (S. Wang & Li, 2022).

As China's governance affairs became more complex, centralized power became difficult to meet the demand. Therefore local government functions began to expand, providing the time for further development of decentralization. Additionally, the allocation of duties between the federal and local governments with regard to pension insurance has been considered (X. Wu, 2011). And it's important to have a clear distinction between central and local government responsibilities in order to support the continued development of the pension insurance system. In order to fix the current unfair responsibility distribution and

eliminate the informational disparity between central and local governments, it is fair and appropriate to separate the administration and financial obligations of central and local governments (Shu, 2021).

3.1.2 Study of government responsibility optimization

The challenges to basic pension insurance in China have increased due to increasing internal and external risks, and therefore the problems faced by government responsibility have become increasingly serious (Lv, 2021). The government did not assume political, legal, and financial responsibilities in the change of rural social pension insurance (Gui & E, 2007). Comparing the performance of government responsibilities in China with that in developed countries and can conclude that the performance of government responsibilities in China is poor, which is reflected in poor quality and lack of comprehensiveness. For example, the government is too centralized, has insufficient fiscal spending, low social activism, inadequate legal system, insufficient supervision, poor management organization efficiency, vague responsibility for underwriting guarantees, lack of scientific and systematic policies, and insufficient management checks and balances (Qi, 2016).

The many challenges to government responsibility in pension insurance are due to a combination of factors. The author argued that the causes of government responsibility are influenced by both subjective reasons of socio-political and economic culture, and objective factors of economic development level and social management system. From a subjective perspective, the lack of government responsibility is due to the government's long-standing neglect of rural social pension insurance. From an objective perspective, it is due to the backwardness of the rural economic development level, the constraints of the urban-rural dualistic social security system, the inadequacy of relevant laws, and the influence of the concept that rural families bear their pensions (Gui & E, 2007). Other authors analyzed the constraints of government responsibility-sharing from five aspects. First, the basic pension insurance system's top-level architecture is flawed. Second, there is an imbalance in the geographical growth of basic pension insurance. Third, there is a lack of desire on the side of local governments to join the pension insurance scheme. Fourth, there is a shortage of reasonable public gambling as well as clear policy standards for transfers between the federal and local governments. Fifth, there aren't enough laws and rules to support pension insurance (Zhu & Zhang, 2017).

The problems in the government responsibility mechanism of pension insurance can bring many negative effects. A decrease in farmers' desire to sign up for old-age insurance, a reduction in old-age insurance welfare, a decrease in rural households' faith in the social welfare and security system, an increase in future fiscal expenses, and an inability to protect rural households' pension rights are some of the disadvantages (Cao, 2011). The risks sustained by the fundamental old-age welfare system will increase as a result of the current issues facing government accountability. For example, the government must bit the gap between income and expenditure to cover the bottom, forming a financial risk; the empty account in the individual account in the pension, accumulating the risk of hidden debt; urban-rural differences, regional differences, and group differences together resulting in the risk of income differentiation; the inability to centralize funds for management and investment, leading to inefficient investment risk; in addition to operational risk, moral risk, etc. However, local government risk liability is reliant on and will eventually be transferred to the national government, which will compromise government risk liability integrity (Lv, 2021).

In terms of the path of optimization, scholars mostly emphasize that the government should actively assume the main responsibility for pension insurance and fill the gap left by the failure of other subjects. The author makes four suggestions for enhancing the government's contribution to the development of this old-age insurance. First, to encourage the creation of proper laws and standards that make clear the primary duty of the government; second, the central government should strengthen the responsibility of top-level design and form an institutional framework that integrates direction and overall situation; third, the government should change the concept of unlimited finance and establish a reasonable and incentive payment mechanism; finally, to improve the service level of operators and institutions and improve the supervision system to promote the efficient and reasonable operation of the system (Q. Yang, 2019). The government responsibility system should be reconstructed, firstly, by strengthening top-level design, secondly, by increasing financial support, thirdly, by efficiently coordinating resources and establishing collaboration and counterpart support mechanisms, and fourthly, by bringing into play the value of land-based and family-based pensions. Additionally, several academics stress the importance of strengthening the system of responsibility-sharing between the federal and municipal governments (X. Yang & Wang, 2019). In order to encourage and restrain local governments, it is also necessary to decide the proportion of the national

government that goes to local governments as well as to improve central-to-local transfer payments (Zhu & Zhang, 2017).

3.2 Study on the willingness of choosing pension insurance

Scholars have mostly analyzed the factors influencing rural residents' willingness to participate in insurance from both internal and external perspectives. Internal factors are mainly divided into personal and family factors. External factors are divided into rural residents' awareness of relevant policies and expectations of the future. To explore the impact on the willingness to join in insurance, the majority of the analysis approaches employ ordered probit models, logit models, OLS regression analysis, and dichotomous logistic regression analysis.

When examining personal traits, researchers frequently start with demographic data and examine how factors like gender, age, marital status, education, occupation, income, the number of children, and physical health may affect one's propensity to purchase insurance.

According to an empirical analysis of field research data from the province of Hubei, male are less likely than female to purchase insurance because they face longer-term retirement risks because of China female's higher average lifespan than male (Deng & Liu, 2013). However, some scholars hold opposing views. Age is also related to the willingness to participate in insurance, for every 10-year decrease in the age of rural residents, the probability of participating in insurance will decrease by 4.2%, and that willingness to take part in insurance is positively correlated with age. This is because as age increases, the retirement problems faced by rural residents become more serious (S. Hu & Shen, 2016). The age of 45 is a significant age cut-off, and residents are much more willing to enroll in insurance after the age of 45 (W. Li & Jiang, 2015). The education level was inversely related to willingness to participate in insurance. Pension insurance is less attractive to rural people with high and medium education, so each year of education decreases the likelihood of participating in insurance by 2.4% (S. Hu & Shen, 2016). Other writers, on the other hand, came to the conclusion that the willingness to take part in insurance is positively correlated with education. This is because rural residents with higher education are better able to understand the content and concept of pension insurance (Y. Xu & Zhang, 2017).

Married rural residents were more willing to participate in insurance compared to unmarried rural residents. Divorced and widowed residents also had a stronger willingness to participate in insurance than those who were married.

Meanwhile, the author found no significant effect of occupation on willingness to participate in insurance (Y. Xu & Zhang, 2017). Income levels are correlated with willingness to. The willingness to join in insurance increases with income level. The willingness to join in insurance will also increase as the number of children rises, and each new child will raise the likelihood by 62% (N. Zhang, 2015). An opposing viewpoint is held by another author, who contends that when income reaches a certain level, people's willingness to engage in insurance will be significantly diminished (W. Huang, 2017). Most elderly rural residents earn their income from physical labour, and as their age increases, the risk of health conditions increases. Health factors directly affect their labor income, so older people with poor health are more willing to insure as a way to reduce the burden on their families (Mu & Yan, 2012).

In terms of household characteristics, scholars mostly analyze factors such as household size, the number of migrant workers, household income status, the area of farmland in the household, and being "bundled" by household insurance. Family pensions can replace social pensions, thus as family sizes grow, people's willingness to take out insurance will also decline. The intention to participate in insurance is strongly connected with the number of families who work not at hometwon because family members who commute will have an adverse effect on the pension protection of other families (Gao, 2012). But there is no direct relationship between per capita household net income and whether rural residents participate in insurance (W. Li & Jiang, 2015). However, other scholars argue that argued that an increase in household per capita income would instead discourage rural residents from participating in insurance. The authors also found that a decline in comparative land returns resulted in a 4.2% increase in the likelihood that inhabitants would enroll in insurance as the amount of agricultural land in the family rose. The "bundled" policy has a favorable link with the propensity to purchase insurance, and it has a considerable impact (S. Hu & Shen, 2016).

Numerous academics have also looked into how external variables affect people's desire to enroll in insurance. It was found that rural residents' level of concern about their future retirement problems significantly influenced their willingness to insure, with each higher level of concern increasing the willingness by 3.044 times (Y. Xu & Zhang, 2017). And the more rural residents favour saving for their old age in terms of future expectations, the stronger their willingness to enrol in insurance. Rural residents' desire to join in insurance was positively correlated with their understanding of urban and rural citizens' pension insurance

policies (W. Huang, 2017). The level of trust that rural inhabitants had in the social pension insurance program had a substantial impact on how willing they were to participate, and as trust rose, so did the level of interest (Y. Li, 2013). Researchers have also discovered that citizens are more motivated to engage in pension insurance when they are satisfied with the policies, institutions, and office employees (S. Hu & Shen, 2016).

In conclusion, rural dwellers have a deep awareness of involvement in social pension insurance and a desire to protect their old age through it, but their willingness to do so depends on a variety of conditions. The findings of academic studies on participants' insurance willingness serve as a guide for creating the questionnaire.

3.3 Study on the influencing factors of payment brackets

3.3.1 Study on the current situation, problems, and causes of payment

By examining the present state of payment levels for this insurance, the author identifies three important contemporary difficulties. First, numerous payment level options can better serve the needs of rural inhabitants, who can select the best option based on their preferences and financial capacity. Second, each local government has modified the range of payment levels and the methodology used to divide them in accordance with the local circumstances, which is more consistent with the growth of the area's economy and the variations in participants' capacity to make contributions. Last but not least, some local governments in each area implement policies to encourage citizens to make larger payments overall and to contribute for longer periods of time (M. Zhang & Sun, 2016).

The government has made various adjustments to the payment model and payment levels for the pension insurance for rural inhabitants as time has changed, especially for the payment levels. The government has also enhanced subsidies and added incentives in the hopes that rural dwellers will become more inclined to pay more. However, a large proportion of rural residents still choose the lowest payment level (Meng & Wang, 2022). Rural dwellers will only symbolically select the lowest level of old-age insurance, despite being under internal and external pressure to do so. This phenomena could significantly limit the ability of the new kind of rural social old-age insurance to provide security (Nie & Zhong, 2014). Researchers discovered further that the payment level of participants did not significantly rise the next year and that most rural inhabitants tended to select the

lowest payment level. Taking the statistics of Yinchuan City, Ningxia in 2018 as an example, about 75% of people chose the lowest payment level and only 3.8% chose the highest purchase level (C. Liu & Chen, 2021).

In Shaanxi Province, a survey of three cities revealed that rural inhabitants' estimated payment levels were concentrated between high and low levels, with 26.6% expecting to pay RMB 100 and 37.1% expecting to spend RMB 1,000–2,000. Less than 10% of rural inhabitants are desired to pay 300-900 yuan, and just 10.3% of those living in rural areas are prepared to invest 200 yuan. This implies that there is a contradiction between the actual and expected contribution levels (Wen et al., 2014). Through the study of five cities in Hebei Province, it was found that about 20.5% of rural residents' expected payment bracket was RMB 100, and about 38.7% of rural residents' expected payment level was RMB 1000-3000. On the other hand, rural residents' actual contributions are centered at RMB 100. This demonstrates that rural people' actual payment level and expected contribution band are substantially different (Y. Li & Liang, 2018).

In addition to the aforementioned issues, researchers have discovered that China's pension insurance system is poorly coordinated and lacks top-level design. This is evident in the stark difference in contribution rates between provinces and municipalities, which will hinder the spread of social justice (Trinh, 2006). In actual operation, the government grants a subsidy of RMB 30 for both the lowest payment level and the higher payment level. And the only way to increase the value of the fund is to deposit it in a bank and earn interest on it. These measures are not conducive to raising the payment levels of residents and do not achieve more contributions and more income (H. Luo, 2014). By examining and classifying the contribution-related policies in some regions of Hubei Province, it was discovered that there is an issue with the subsidy policy's extreme regional variation, which makes it more challenging to manage the system. The way of granting additional subsidies against the basic pension is also unreasonable. At the same time, most counties have adopted the method of increasing contribution subsidies by absolute amounts, and this method is not in line with the choice behaviour of rational economists (Xue et al., 2014).

In essence, social pension insurance is a government initiative designed to better safeguard the senior population's vital means of subsistence. However, there are further issues with payments, which will substantially lower the policy's effectiveness. Numerous scholars have explored the reasons for these problems to find solutions. The limitation of individual income level is an important reason.

Although people's income levels have increased with economic development, consumer spending has also increased significantly, so pension insurance costs still put some pressure on family finances. The traditional concept of old-age pension also affects the contribution, and the elderly prefer to rely on their children rather than the government. There are also loopholes in the government's policy dissemination, and residents are not motivated to learn about the policies (Meng & Wang, 2022). Chinese people's realistic view of benefits makes people prefer to control their savings, and the pay-out period of pension insurance is too long. Grassroots social pension insurance administrators are obsessed with coverage and do not guide residents to choose the most appropriate contribution bracket. The herding effect also makes a large number of residents follow the trend of choosing the lowest payment level (H. Luo, 2014).

3.3.2 Study on the behavior of choosing payment brackets

The focus between the two is different, and there are differences in both whether a resident participates in the urban and rural citizens pension insurance and the contribution level they choose. Residents won't have to decide on a contribution level until they elect to enroll in the urban and rural citizens pension insurance. The choice of contribution level also reflects the citizens' understanding of pension insurance. Residents' choice of contribution level is highly subjective, and many factors influence this behaviour. At present, many scholars analyze the factors influencing the behaviour of contribution grades based on household surveys and field research. These research results can be summarized into two aspects: on the one hand, the analysis of factors affecting the contribution level, mainly from the participants' situation, analyzing personal characteristics, family characteristics, policy cognition, economic income status, etc.; on the other hand, this is for the problems existing in the social security policy system, analyzing the expected protection benefits, contribution incentive mechanism, policy publicity intensity, etc.

The logistic model was created using the stepwise regression technique using data from Yinchuan City individuals who took part in pension insurance in both urban and rural areas. The author found that age was positively correlated with the level of payment. Younger insured individuals are more inclined to select a lower premium tier. Education level and premium level have a negative relationship. This indicates that people are more likely to select a larger premium the less educated they are (C. Liu & Chen, 2021). The author discovered that residents with higher

incomes will choose the higher level of payment through a typical sample survey of rural residents in five cities of Hebei Province using a logistic model. The author also discovered that the level of education has a meaningful influence on the level of payment. However, the author believes that the lower the education level, the more likely the residents are to choose the lower payment grade (Y. Li & Llang, 2018).

Based on research data from Baoji, Xi'an, and Tongchuan cities in Shaanxi Province, using multiple regression and ANOVA, the author found no significant differences in the contribution brackets chosen by residents aged 18-28, 29-39, and 51-59 years old. Residents of various marital statuses behaved much differently when selecting their contribution brackets, nevertheless. Widows are likely to choose the lowest contribution bracket than married and unmarried people. This is because widowed households have less labour force, which will directly affect the household's financial strength to afford higher contribution levels (Wen et al., 2014). Also, occupation affects the behaviour of participants in choosing contribution levels. The author analyzed a sample of urban farmers in Chengdu city and found that in the highest contribution bracket, more participants were not engaged in agricultural production activities than those who were engaged in agricultural production activities. In the lowest contribution bracket, the opposite is true, with 31.7% of the participants not engaged in agricultural production activities. This is because being engaged in agricultural production activities has a high probability of influencing the participants' economic strength, knowledge of social policies, awareness of pension insurance, and perception of pension, which further influences the behaviour of the contribution bracket (Cheng, 2014).

The author discovered that gender had no discernible influence on the decision of which contribution bracket to choose when conducting field research in Henan Province using the interaction approach and logistic regression method. Most of the male and female participants choose the lowest contribution bracket. Additionally, it is discovered that age has no bearing on the choice of contribution bracket by contrasting the group of those under 45 with those over 45. Most frequently, both groups select the lowest contribution tier. Physical health also has less impact on the contribution bracket, and regardless of their health status, participants still mostly choose the lowest contribution bracket (Lin, 2017). Using the hierarchy of needs and rational choice theory to analyze the behaviour of Changshu residents in choosing the contribution level, the author found that the older the age, the more residents would choose a higher contribution level. Age and

physical health are strongly linked. In general, as people get older, their health deteriorates more and they become more concerned about the future pension issue. Therefore, the poorer the health condition, the more residents tend to choose higher contribution levels. The author also discovered that because countryside individuals typically lack access to higher education and are therefore less knowledgeable about old-age insurance policies, they tend to choose lesser contribution bracket (X. Xiao et al., 2022).

Some scholars also put forward that there is a U-shaped relationship between residents' choice of payment level and their age. That is, the older and younger residents tend to pay a higher level of payment, while the median residents tend to pay a lower level of payment (Deng & Li, 2014). Other scholars put forward that with the development of China's economy and society, the personal factors of insurance enrollees no longer affect their choice of different payment grades as in the past. On the one hand, the pension insurance system's promotion frees the insured from cultural limitations. However, as people's income levels rise, inhabitants of both urban and rural areas may comfortably pay the annual pension insurance cost of several hundred yuan (H. Luo, 2014).

Hypothesis 1: The choice of this old-age insurance categories for rural people is influenced by the demographic characteristics.

The author discovers a positive correlation between annual household income and contribution grade using data from a field survey conducted in the provinces of Jiangsu and Zhejiang using the ordered probit model. The annual household income reflects the living standard of the household. At the same time, households are more affluent and therefore more willing to pursue welfare needs such as pensions and have a stronger willingness to make contributions and choose higher contribution levels. However, the author also found that this result is not robust, probably because it is also influenced by the size of the household and the degree of awareness of the individual (Deng & Li, 2014). Based on the analysis of a sample of urban farm households in Chengdu city, it was found that as the net household income per capita increased, participants were likely to choose a higher contribution bracket. There is a correlation between per capita household income and the choice of contribution level, but it is not a simple positive correlation. However, as per capita household income increases further, the participants' demand for pension insurance becomes less intense, so the tendency to choose the highest contribution bracket decreases. The choice of net household income per capita and contribution bracket showed a U-shaped relationship (Cheng, 2014).

By conducting a field survey in 10 rural-related districts and counties in Xi'an, Shaanxi Province, and using multi-categorical logistic regression analysis, the author found that annual basic household expenditure was also significantly associated with the choice of contribution bracket. The annual basic household expenditure accounts for a relatively large proportion of total household expenditure, especially among low- and middle-income households. The amount of disposable income that a household has available will decrease if its basic expenses increase during a time of steady income. As a result, families with higher basic needs tend to choose contributions with lower amounts (Feng & Dong, 2010). The author discovered that traditional agricultural production methods have an impact on rural areas and that most parents continue to live with their children after losing their ability to work through a survey of farmers' insurance participation in regions with various levels of economic development. The traditional idea of a family pension has a long history. Since the family pension has a substitution effect on the protection function of this old-age insurance, local dwellers will choose the lowest contribution class to cut expenditures (M. Zhang & Sun, 2016).

Based on an empirical study in ten cities in Shaanxi Province, using a binary logistic regression model, the author explores the effect of child structure on participants' choice of contribution bracket. The number of children significantly influences the participants' choice of contribution bracket, and only-child families are likely to choose a higher contribution bracket. The probability of choosing the lowest contribution bracket for one-child families is 44.8% of that for other multi-child families. The author further analyzed the effect of having daughters on the choice of contribution bracket and found that participants without daughters were likely to choose a higher contribution bracket. The probability of choosing the lowest contribution bracket for families without daughters is 47.8% of that for families with daughters. And as the number of daughters increased, the proportion of choosing the lowest contribution bracket also increased significantly. However, the presence or absence of sons does not significantly affect participants' choice of contribution bracket selection (Wang Q. & Guo, 2016).

Hypothesis 2: The choice of insurance brackets made by rural dwellers is influenced by the household characteristics.

Through field research in twelve rural areas of China and using an ordered Probit model, the author finds that farmers' knowledge of pension insurance policies significantly affects contribution grade selection behaviour. Residents with more in-depth understanding of old-age insurance policies are more inclined to use

public old-age insurance as a retirement alternative. They have a strong willingness to pay contributions and have a sense of worry and are willing to prepare for their retirement in advance and choose a higher contribution level (Wang L. & Liu, 2015). By analyzing the behaviour of rural residents in Hebei province in choosing their contribution brackets, the author also believes that the higher the residents' awareness of the policy, the more they will choose higher contribution brackets. Unfortunately, in the field study, it was found that most rural residents have low awareness of pension insurance and only know the amount of basic pension. Even most of the village cadres and grassroots administrators have low awareness of pension insurance (Wu C., 2015).

The author builds two-category and multi-category ordered logistic models using survey data from 393 insured farmers in Anhui Province, and discovers that there is a "knowledge gap" effect in farmers' access to pension insurance policies, which has a deeper impact on the choice of contribution level. Compared with the stronger rural residents, the weaker rural residents are less able to receive policy information, resulting in their low awareness of pension insurance policies. Rural residents choose the lowest contribution level not because they cannot afford a higher contribution level, but because of the "knowledge gap" effect (Deng & Jiang, 2014). The author found that participants were more likely to select a higher contribution bracket when they had a general understanding of this old-age insurance by analyzing survey data from 464 farmers in Junan County using an ordered logistic regression model. Furthermore, the location's publicity has an impact on how much people know about the policy (X. Xu & Xue, 2016).

Individuals' perception of their economic power also significantly influences the choice of contribution bracket. Based on a questionnaire survey of rural residents in four provinces in China, the author found that about 60% of rural residents chose the lowest contribution bracket because they thought they could only afford the lowest contribution bracket based on their economic power. However, the reality is that most rural residents can afford higher contribution levels (Nie & Zhong, 2014).

Hypothesis 3: The selection of this old-age insurance brackets by rural dwellers is influenced by policy and subjective perspectives.

The author utilized a logistic regression model to analyze data from approximately 500 households in a prefecture-level city in Henan Province and discovered a substantial positive association between inhabitants' happiness with the service provider and their selection of contribution class. It was found that

when residents' satisfaction with the service provider increased by one level, the number of residents who chose a contribution level of 200 yuan or more increased by 1.344 times. This means that the higher the satisfaction with the service provider, the higher the contribution bracket chosen. The author also found that satisfaction with the policy design also significantly influenced the choice of contribution bracket. When residents' satisfaction with the policy implementation increases by one level, the number of residents choosing a contribution bracket of 200 yuan or more increases by 1.303 times. This implies that the higher the satisfaction with the policy design, the higher the contribution bracket is chosen (Du, 2021). According to the level of economic growth, the author performed field research in four counties in Anhui Province and discovered that system satisfaction affects people's contribution behavior and that citizens are more likely to contribute when they are more satisfied with the policy (Luo X., 2012).

Hypothesis 4: The policy and institutional satisfactions have an impact on rural residents' choice of urban and rural residential insurance brackets.

The psychological expectations of rural residents influence the behaviour of contribution bracket selection. Participants' expectation assessment is influenced by the credibility of the grassroots organization and cadres of the policy implementation agency. Especially in rural areas, the promotion of social pension insurance mainly relies on grassroots cadres, so the impression and credibility of grassroots cadres among participants are closely related to the implementation effect. However, because of historical problems, the service consciousness of grassroots cadres is weak and there are some corruption problems, so rural residents' trust in grassroots organizations and grassroots cadres is low. At the same time, some grassroots cadres pursue high participation rates and other face-saving projects, which makes rural residents rebel against them. As a result of distrust and rebellious psychology, the rural residents' psychological expectation of pension insurance is greatly reduced, which is not conducive to choosing a higher level of contribution (Wu J., 2011).

A study of 400 rural households in Changwu County, Fuxin City, Liaoning Province, revealed that rural residents are generally less educated and therefore do not have better information integration skills, and have difficulty understanding and judging new policies. Therefore, most of the rural residents did not start from their actual situation, but were influenced by the people around them and had a "herd mentality". When they are aware that those close to them selected the lowest contribution bracket, they do the same (Lu H., 2012). By analyzing field research

data from four counties in Henan Province, the author found that rural residents could not judge new things, and the publicity of the pension insurance policy did not meet expectations, so farmers passively chose the contribution level. It is mostly affected by village cadres, prestigious village citizens, and wealthier village residents, creating an issue of blind obedience between villages, between farmers and farmers, and between village residents and village cadres (Zhang H. & Liu, 2013).

Hypothesis 5: Rural residents' decisions on urban and rural residential insurance brackets are influenced by external factors.

In summary, the economic base, the current state of policy implementation, local policies, and the socio-cultural environment of retirement all differ significantly because of the different research regions selected by the researchers. The conclusions drawn on this basis have a large difference. Therefore, in the follow-up study, field visits and surveys must be conducted in the selected research areas to achieve a specific analysis of specific problems.

3.4 Study on the promotion of payment brackets

3.4.1 Research on the evaluation of pension insurance

Researchers have examined the implementation impact of pension insurance programs for urban and rural inhabitants from many angles, but no consistent result has been reached because of the varied attention paid to the issue and the data used.

According to some experts, the adoption of this old-age insurance program is noteworthy. At this point, the algorithm takes into account the actual national conditions in China. Additionally, it comprises rural citizens who are economically weak in the system and those in flexible employment in urban areas. It follows an equal approach in the payment of fundamental pension standards and does not require residents to pay contributions or to choose the grade of contributions. These exhibit the idea of justice and fairness. And this approach alleviates the financial hardship experienced by certain older persons in rural and metropolitan areas (Z. Wang, 2017). This system's construction also heralds the launch of China's main social security system, which is crucial for advancing social justice and equity for both urban and rural citizens. Additionally, the division between metropolitan and rural areas has been eliminated, enabling the equality of fundamental government services and the early satisfaction of the Chinese people's demand for

"a sense of security in old age". Despite the fact that more work needs to be done in order to reach higher levels of pension payments, a fundamental shift from nothing to something has been made. Additionally, as the policy has been promoted, insurance coverage has progressively increased, and the level of pension benefits has also risen year over year (Y. Yang, 2018).

Some researchers also think that the old-age program's adoption was impacted by a number of reasons and that it hasn't been entirely successful. There are still problems of "adverse selection", such as low motivation of young people to participate in the insurance and the concentration of participation in the lowest contribution bracket. A complete solution to the pension issue is thus a fantasy due to the low replacement rate of personal accounts in the future and their inability to guarantee security for the lives of the elderly (C. Xiao & Liu, 2018). According to an analysis of the panel three-stage DEA model, the operational efficiency of this old-age insurance program needs to be increased. There are also significant regional differences, with the eastern region having the greatest operational efficiency and the central region having the lowest operational efficiency (Y. Hu & Li, 2019). By analyzing panel data from 31 provinces between 2013 and 2019, the author found that the overall performance of this old-age insurance scheme has declined due to the constraints of technological innovation. 21 provinces did not reach DEA validity in 2019 (Yin & Zhong, 2021).

This old-age insurance program's original intent was to unite urban and countryside old-age security into one system, end their historical isolation, and advance fairness, justice, and coordinated growth. However, numerous issues have come to light during the deployment of this system. For instance, the location of the pension insurance scheme is ambiguous, the degree of basic pension security is poor, people typically make modest levels of contribution, and the government agencies' ability to provide services is limited. All of these issues limit the ability to significantly strengthen the pension insurance programs for urban and rural populations (Ge, 2021). The efficient operation of the pension insurance mechanism has been impacted by both aging of the population and rapid economic development. With the acceleration of urbanization, the living standard of rural residents has increased, but there has been a stagnation in the scope of work and fundraising, and the related pension insurance system does not conform to the law of development and supporting policies have not been introduced. And the publicity is not enough, which makes the rural residents' understanding of pension insurance low. The government's centralized management of funds is insufficient,

supervision is not in place, and interest rates are low. In addition, the coverage rate has not reached the desired target, and the low amount of participation and the weak renewal ability also impact the system operation (Qiong & Yan, 2007).

3.4.2 Research on promotion of the payment mechanism

Researchers have looked at the mechanism for increasing pension insurance contributions from urban and rural inhabitants from a variety of angles and disciplines. The choice of whether to enroll in the insurance is made in the context of voluntarism, despite the fact that the old-age insurance pilot is still in its early phases. However, if we refer to Germany, the United States, and other developed countries, we can find that the essential feature of their basic social pension insurance is compulsory. China's pension insurance system is also a public-assisted self-insurance type of social pension insurance in essence, which should follow the compulsory principle of social insurance. In addition, in order to address the issue of reverse involvement in the voluntary scenario, the government must take advantage of the chance to switch from "voluntary participation" to "compulsory participation" since farmers' incomes have increased (Zhao & Hai, 2013). The primary aim of this old-age insurance, however, is the unemployed citizens because they earn less than other categories. Therefore, after China achieves its 2020 aim of entirely eliminating poverty and building a society that is moderately wealthy, these old-age insurance payments will become obligatory payments. And concerning a more reasonable range of contribution standards and residents' ability to contribute as such, set the contribution brackets and choose an appropriate time to change to a proportional rate system (Zhang Y. & Xue, 2017).

The government's financial assistance ensures the long-term viability of the pension insurance program for urban and rural citizens. However, the existing financial incentive mechanism, due to the low crediting rate and the lack of a progressive individual contribution financial subsidy mechanism, cannot motivate residents to choose a higher contribution level that better suits their own pension needs, but only to pay insurance earlier or for a longer period. If taking a holistic approach, the financial incentive mechanism should have an income redistribution effect, but some of the subsidy criteria apply the principle of absolute fairness and therefore lose the original income redistribution effect (Liu, H. Y., 2016). Based on the partial equilibrium analysis framework, the incentive effect is studied by actuarial methods, and the incentive effect of fiscal policy is also found to be poor. On the one side, this is due to the fiscal approach to subsidies has only partial

intergenerational redistribution effects but no intra-generational redistribution effects. However, despite the fact that the government provides subsidies to farmers through the Treasury, there is also a devaluation of personal accounts, which has an adverse effect on the banks. Therefore, the incentive effect of the fiscal subsidy is only for middle-aged and elderly farmers and does not motivate residents to choose higher contribution levels nor does it motivate young residents to purchase insurance. If the financial subsidy is further increased, it will generate a retrograde transfer of income and put the government in a difficult situation of heavy financial burden (Chen, Yunfan & Tan, Xuan, 2015).

China's social pension insurance system is still in its infancy and can learn from the valuable historical experience of western countries. Western countries believe that under the premise of guaranteeing the basic livelihood of citizens, the axis of interest incentive function of social security should also be brought into play. First, residents should be incentivized to increase their share of the social security fund supply. Residents should be explicitly informed that the share they receive of social security funds is directly related to the social security funds they pay, to increase their motivation to pay social security contributions. Second, establish a reasonable distribution mechanism for social security benefits, where wages are linked to the social security contributions paid, as well as to allowances and security contributions. This can build a favourable atmosphere of paying more for more work and receiving more for more contributions (Han, 2016). In addition to linking income to the level of contributions, the fund return rate should also be increased, and contributions should be linked to contribution subsidies to build a suitable adjustment mechanism for the basic pension (Liu Z. et al., 2016). After evaluating a number of concerns presented by the integration of this old-age insurance, one of the more important issues is the ineffective effect of the incentive mechanism of the old-age insurance program. Therefore, the incentive mechanism can be made more specific by gradually grading the integration, increasing the level of integration of pension insurance, strengthening the operational capacity of relevant staff, etc. (Yang D., 2016).

Incentives can only work if residents' level of policy awareness is raised. The existing advocacy efforts are small and rely mostly on group meetings. As a result, on the one hand, it's important to choose a talented team of employees as as quickly as possible and establish a dedicated department for policy propaganda. Each employee's professional knowledge should be increased, and propaganda efforts should be carried out in cities, villages, and communities to increase awareness

among locals of the pension insurance available to both urban and rural citizens. On the other hand, it's important to make residents aware of and understand issues like contribution amounts, government subsidy laws, and pension payments that are practically important to them. It is also important to expand the publicity methods, such as filming short videos, making cartoon books, cell phone text messages, and broadcasts, to increase the residents' motivation to participate in the insurance and renew the insurance (Zhou, 2016). For residents under 45, it's crucial to intensify marketing efforts and use them to raise their awareness of insurance enrollment. It also helps residents to better calculate the economic accounts of contributions and understand that the earlier they participate in pension insurance, the higher the social pension insurance benefits they receive (Li X., 2013).

Participation in urban and rural inhabitants pension insurance is also influenced by the government departments' ability to provide services. In order for citizens to receive effective and high-quality services, it is vital to increase the handling capacity of government agencies and the level of service. Firstly, simplify the handling process based on the principles of convenience and efficiency, observing safety and risk prevention; secondly, make good use of the advantages of Internet development to improve the level of service informatization, introduce online payment and online application for benefits, and help improve the complex problems in the public service of social insurance; finally, in order to successfully care for the aged population who do not have access to the Internet or the physically challenged, the government can establish representative positions at the local level (Wang C., 2021). By hiring NGOs to provide public services, the government can also serve as an effective role model. NGOs can play a valuable role in educating and encouraging individuals to sign up for this pension, changing the social security herd mentality. The participation of NGOs also strengthens the service team, which can provide residents with cleaner and more efficient pension services and cater to the diversified and specialized needs of pension services in today's society (Lin, 2017).

The government must seek to incorporate this old-age insurance into the integrated development of metropolitan and rural areas based on the invaluable experience obtained from the integration of old-age insurance in Chengdu. The government should establish an effective pension insurance transfer policy, raise the level of coordination and strengthen the level of information technology. Additionally, it must encourage the integration process and end the separation of

identities between urban and rural populations. Every reform should be centered on creating a single pension insurance system for citizens, which requires top-level knowledge and concentration on system design (Wang H., 2017). Considering that this old-age insurance offers less protection than the pension insurance for urban laborers. Younger residents and some higher-income rural villagers prefer the latter. Due to their high rate of job mobility, these groups find it challenging to pay for adequate pension insurance while they are looking for work. In order to facilitate the transition between the urban workers' pension insurance system and the rural and urban inhabitants' pension insurance system, rules and guidelines must be introduced. This will make it easier for these groups to choose the right insurance under different employment statuses, and thus increase their motivation to pay contributions (Wang J., 2021).

3.5 Summary of the literature

It has been discovered through a review of the literature that China has more studies on urban and rural inhabitants pension insurance than other nations do. Chinese scholars have made a lot of studies on the factors influencing the intention to participate and the factors influencing the contribution level. However, almost no research has been conducted in other countries on the factors influencing payment levels. Instead, they focus on the division of responsibility of the main contributors, fund supervision, and system implementation.

There are, however, less studies on payment levels in China because academics there favor researching insurance willingness. In addition, there are fewer research on this old-age insurance program than there are on new kind of rural social old-age insurance due to the late commencement of this program in China. And most of the existing studies take national or provincial as the research perspective, which lacks the regional aspect and is not suitable to be applied to specific regions.

Using this information, this manuscript chooses Teng Zhou City in Shandong Province as the study object. It then gathers field research data and constructs a model. In Teng Zhou city, the factors influencing the amount of pension insurance payments to urban and rural inhabitants are examined in terms of individual, household, and policy perspectives. And based on the findings, suggestions are made to help improve the current more imbalanced choice of payment levels.

4 Methodology

4.1 Research design

The installation of this pension insurance has resulted in an imbalance in the contribution bands. This thesis initially investigates the variables that affect how rural inhabitants behave while selecting their contribution bracket in order to remedy this issue. It then analyses the direction of each factor's influence on contribution brackets. Finally, corresponding recommendations are made based on the conclusions drawn.

A questionnaire was administered to collect the respondents' choice of contribution bracket, the influencing factors hypothesised above, and other supplementary data. Once first-hand information was obtained, the data was first collected and collated, errors or omissions were eliminated, and descriptive statistical analysis of the variables was conducted. Next, using quantitative research methods, appropriate models were set up to analyse which factors influenced contribution brackets. Finally, based on the literature collated, the findings of the data analysis are combined to provide recommendations for contribution bracket promotion in the study area.

4.2 Research site

Tengzhou is a county-level city in Shandong Province, China, under the administration of Zaozhuang City, a fourth-tier city in China. In 2021, urban residents' per capita disposable income will be about twice as high as rural residents' per capita disposable income.

Tab.2 Number and composition of the household population in Tengzhou at the end of 2021

Indicator	Year-end number (persons)	percentage
Total population	1764971	100.00%
Among them: urban	891251	50.50%
Rural	873720	49.50%

Among them: Male	931521	52.80%
Female	833450	47.20%
Among them: 0-17 years old	393851	22.30%
18-34 years old	385495	21.80%
35-59 years old	632723	35.90%
Age 60 and above	352902	20.00%

Source: Tengzhou Bureau of Statistics, 2022.

The aforementioned table shows that Tengzhou has a total population of 1.765 million, with slightly more men living there than women. And the problem of ageing is not encouraging, with the elderly population being roughly the same as the youth population.

4.3 Sampling method

The data used in this thesis were obtained from field research conducted by the author between June and September 2019 among rural residents in the Tengzhou City area of Shandong Province. The research was conducted in 10 of the 17 townships under the jurisdiction of Tengzhou City. The 17 townships under the jurisdiction of Tengzhou City were divided into near-urban townships (A-type townships) and far-urban townships (B-type townships) according to their proximity to the main urban area, and a stratified random sampling method was adopted to randomly select three townships from the five A-type townships, namely, Dongshahe, Hongxu and Jiangtun, and seven townships from the 13 B-type townships, namely, Binhu, Jiehe, Xigang, Zhangwang, Dongguo, Yangzhuang and Chaihudian. Through random household surveys and random intercept interviews at marketplaces, a total of 908 people were surveyed and 908 questionnaires were formed. As this study focused on the factors affecting the level of rural residents' basic pension insurance contributions, the sample aged 20-59 was retained, while those who did not participate in this insurance were removed. Through data cleaning and sorting, the final sample size for this thesis's research was determined to be 800.

5 Data and model

5.1 Payment brackets of pension insurance in Tengzhou

According to the real poll, there were only 4 respondents in the RMB1,000 and above category. The difference between the contribution brackets was small, so the contribution brackets were reclassified.

Tab.3 Current status of urban and rural inhabitants' old-age insurance contribution brackets in Tengzhou

Payment grade	Instructions	Meaning	Frequency	Percentage (%)
1	300 RMB and below	lower	440	55.00
2	500 RMB	low	197	24.63
3	600 RMB	ordinary	72	9.00
4	800 RMB	high	58	7.25
5	300 RMB and below	higher	33	4.13

More than half of the sample's participants, as shown in Tab.3, chose bracket 1, which is where the majority of respondents fell. And as the contribution bracket increases, the number of participants decreases. The number of participants in the highest contribution bracket only accounts for 4.13%. This shows that Tengzhou's contribution bracket for rural and urban inhabitants' pension insurance is low, indicating that efforts must be made to increase the desire of rural inhabitants to participate.

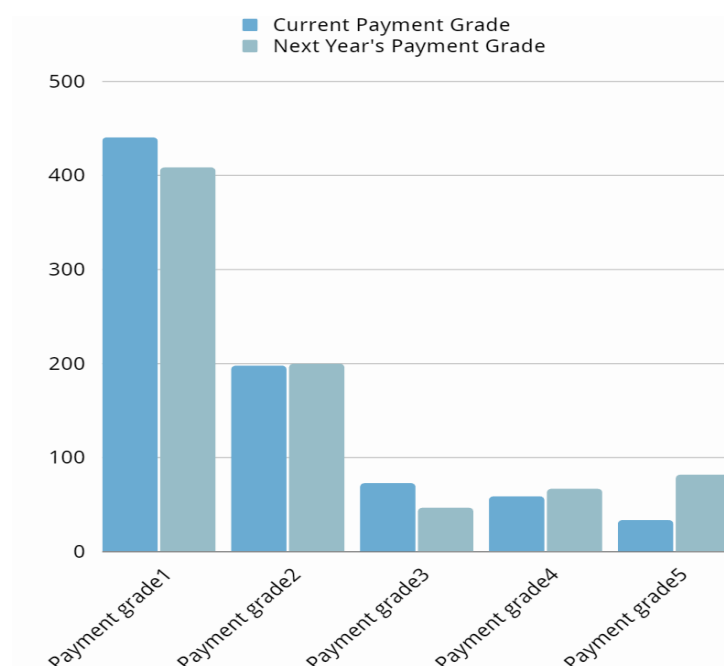


Fig.5 Comparing the quantity of participants in the currently chosen contribution category with the quantity of participants in the following year's

Figure 5 demonstrates that, in comparison to their current choice of contribution bracket, fewer people will select bracket 1 in the coming year, while more will select bracket 5. There has also been a decrease in the number of choosing bracket 3 and an overall optimisation of bracket. However, there is little change and there is inertia in the choice of contribution bracket, as people habitually choose their old contribution bracket without external factors. The pension insurance still suffers from the dilemma of low contribution levels, which does not stimulate its vitality. Therefore, must break the inertia and promote a higher contribution bracket for rural residents through multiple channels.

5.2 Demographic characteristics of rural residents

The demographic characteristics of rural people are described in this thesis and may be found in Tab.4. These characteristics include gender, age, marital status, occupation, level of education, and state of health.

Tab.4 Statistical description of demographic characteristics of rural inhabitants

Variable	Description	Mean	Std. Dev	Min	Max
Gender	Female is 0; male is 1	0.521	0.500	0	1

Tab.4 (continued)

Variable	Description	Mean	Std. Dev	Min	Max
Age	Actual age of the respondent	45.168	9.392	20	59
Marital status	Widowed is 0, non-widowed is 1	0.940	0.238	0	1
Occupation	In agricultural activities 0, not in agricultural activities 1	0.753	0.432	0	1
Education					
Primary school and below	Primary school and below is 1, no is 0	0.414	0.493	0	1
Junior high school	Junior high is 1, no is 0	0.279	0.449	0	1
High school	High school is 1, no is 0	0.208	0.406	0	1
College education or above	College education or above is 1, no is 0	0.100	0.300	0	1
Health status	Very poor is 1, relatively poor is 2, average is 3, good is 4, very good is 5	3.541	0.861	1	5

The results of a statistical study of the pertinent variables showed that the gender mean value was 0.521., which is largely in line with the male to female ratio in Tengzhou in 2018. The sample's average age is 45.168, which fully demonstrates the older age level of the participants. This could be because some young people go to work and contribute to the basic urban worker pension insurance at their location of employment. In terms of marital status, the mean value is 0.940, which means that most rural inhabitants are in stable marriages. In terms of occupation, the mean value is 0.753, which implies that the vast majority of people living in rural areas do not work only in agriculture.

Tab.4 also confirms that the education of rural residents is generally low, with 41.38% of individuals having primary school education or less, 27.88% having middle school education, 20.75% having high school education; and only 10.00% having college education or above. This is consistent with the reality that China's long-term rural education resources are inadequate, while those from rural areas

who are more likely to qualify for the old-age insurance for urban laborers are those who have finished more formal education. In terms of health status, the mean value is 3.541, which means that rural residents are optimistic about their health status.

5.3 Household characteristics of rural residents

In the Tab. 5 of this study, the household characteristics of rural people are analyzed and described in regard to the number of children, household size, annual per capita household income, and total household assets.

Tab.5 Statistical description of household characteristics of rural inhabitants

Variable	Description	Mean	Std. Dev	Min	Max
Number of children	Number of children	1.539	0.703	0	4
Household size	Household size	4.855	1.230	1	13
Log annual per capita household income	Log annual household income per capita	9.716	0.565	5.441	11.849
Total household assets	1 for RMB 0-0.2million, 2 for 0.2-0.4million, 3 for 0.4-0.6million, 4 for 0.6-0.8million, and 5 for 0.8million and above	2.504	1.246	1	5

According to the table above, the respondents were largely between the ages of 41 and 50, a demographic that is impacted by state family planning. As a result, most rural residents only have 1-2 children, with a highest of 4 children. The mean value for the number of children is thus 1.539, as can be seen. Total household members are represented by a mean value of 4.855, which means that rural areas tend to have small families. Rural residents with annual per capita household income of RMB 0-10,000 RMB 10,001-20,000 and greater than RMB 20,000 accounted for 13.75%, 60.50% and 25.75% of the entire sample respectively. This shows that the majority of rural residents have a medium level of income.

5.4 Policy and subjective perceptions

The study discovered that there are five key channels via which rural inhabitants might learn about the pension insurance policies that apply to both urban and rural citizens. With the development of new media, the influence of newspapers is decreasing, and the Fig.6 shows that few rural residents learn about pension insurance policies from newspapers. It is also found that village cadres and the mobile phone network are the most important channels for residents to learn about information. Consider starting with the most effective channels in our recommendations in order to better advertise and publicize this old-age insurance.

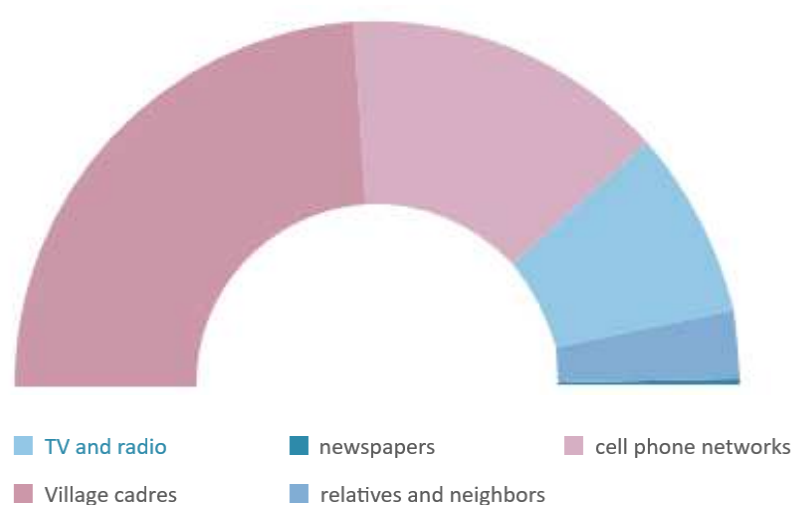


Fig.6 Access to insurance content for residents

This study describes the policy and subjective perceptions of rural residents in terms of their understanding of the content of pension insurance, the amount of future pensions they will receive, the policy of paying more and longer will get more, the destination and operation of funds, government subsidies and the change procedures, which can be seen in the Tab.6.

Tab.6 Statistical description of rural inhabitants' policies and subjective perceptions

Variable	Description	Mean	Std. Dev	Min	Max
Content of pension insurance	Very poorly understood is	2.508	0.928	1	5
Future pension amount	1; very well understood is	2.460	0.935	1	5
Policy of paying more and longer will get more		2.963	1.042	1	5

Destination and operation of funds	5, and scored between 1 and 5	1.993	0.745	1	5
Government subsidies	5	2.105	0.900	1	4
Change procedures		2.373	0.792	1	4

In terms of the understanding of the content of pension insurance, the mean value is 2.508, which means that residents know less about the content of insurance. Residents also know less about the amount of pension they will receive in the future, the policy of paying more and longer will get more, the government subsidies and the procedures for changing insurance. The mean value for the destination and operation of funds is only 1.993, which means that residents do not understand the destination of pension funds and how they operate.

5.5 Policy and institutional satisfactions

This study covers rural people' satisfaction with policies and institutions in terms of grassroots government's operations, operational capability, publicity efficacy, contribution bracket setting, incentive policy, and basic pension payment, which can be found in the Tab.7.

Tab.7 Statistical description of the satisfactions of rural inhabitants with policies and institutions

Variable	Description	Mean	Std. Dev	Min	Max
The work of the grassroots government	Very dissatisfied	3.405	0.736	1	5
Ability to conduct operations	is 1; very	2.981	0.758	1	5
Effectiveness of publicity	satisfied is	2.748	0.890	1	5
The setting of contribution brackets	5,between 1-5 for scoring	3.118	0.745	1	5
Incentive policy		3.343	1.013	1	5
Basic pension payment		2.381	0.916	1	5

In terms of rural residents' satisfaction with policies and institutions, it was found that the mean values of satisfaction with the work of the grassroots government, the setting of contribution brackets and the incentive policies all exceeded 3. This indicates that inhabitants are happier with them. The mean value for the ability to

conduct operations is 2.981, which means that residents are generally satisfied with it. The effectiveness of publicity was also average, but the lowest is the basic pension payment.

5.6 External factors

This study examines the impact of outside variables on rural residents' decision on their contribution level from four angles: family, relatives and friends, village cadres, and new rural cooperative medical insurance, which is available in the Tab.8.

Tab.8 Statistical results of the degree of influence by external factors

Variable	Description	Mean	Std. Dev	Min	Max
Family	Not affected at all is 1;	2.508	0.825	1	4
Relatives and friends	more or less affected is	2.080	0.803	1	4
Village cadres	2; greatly affected is 3;	1.944	0.741	1	4
New rural cooperative medical insurance	completely affected is 4	1.568	0.655	1	4

According to the Tab.8, the mean value of the factor influenced by family members is 2.508, which indicates that family members influence the contribution bracket to a greater extent. The mean value of the factor influenced by relatives and friends is 2.080, indicating that relatives and friends influence contribution bracket to a certain extent. The mean value of the factor influenced by village cadres was 1.944, indicating that village cadres also influenced contribution bracket. The new rural cooperative medical insurance has the lowest mean value of any factor, 1.568, indicating that this factor is the least influential of the four, but still influences. All in all, residents' contribution behaviour is influenced by four of them. Due to China's strong traditional family values and the fact that it is a relational country, homes are affected by both social and familial interactions.

Hypothesis 5: External factors have an impact on rural residents' choice of urban and rural residential insurance bracket, accepted.

5.7 Model design

The explanatory variable in this study is the contribution bracket. This variable is re-designed into five classes according to the participation status, assigned the values "1-5" and given the meanings "lower, low, average, high, higher". The explanatory variable is a typical multi-valued ranking variable and given the characteristics of the variable, an ordered logistic model could have been used, but the model did not pass the parallelism test and therefore the model was no longer appropriate. The generalised Hausman test proved the Independence from Irrelevant Alternatives (IIA) and therefore a multi-logit model was chosen for the analysis.

The explanatory variables are divided into demographic characteristics, household characteristics, policy and subjective perceptions, and policy and institutional satisfaction. Demographic characteristics include gender, age, marital, occupation, education, and health. Household characteristics include the number of children, household size, annual per capita household income and total household assets. Policy and subjective perceptions include understanding of the content of insurance, the amount of future pensions, the policy of paying more and longer will get more, the destination and operation of funds, government subsidies and change procedures. Satisfaction with policies and institutions includes satisfaction with the work of the grassroots government, the ability to conduct operations, the effectiveness of publicity, the setting of brackets, the incentive policies, the basic pension payment.

A multi-logit model can be considered as a joint estimation of several binary logit models with two pairs of each type of selection behaviour in the explanatory variables. The basic expression of a multi-logit is as follows:

$$\ln \left(\frac{\pi_{ij}}{\pi_{ib}} \right) = \ln \left(\frac{P(y_i=j/x)}{P(y_i=b/x)} \right) = x'_i \beta_j \quad (1)$$

Where b is the selected base group and set j is the total number of species contained in the category variable. When j=b, the left-hand side of the equation is $\ln(1/1)$, then $\beta_b = 0$. That is, the log-odds of a given choice are always 0 relative to itself, so that any explanatory variable coefficient corresponding to that group must also be 0.

In this study the selected base group is the contribution bracket 1 and the total number of species contained in the category variable is 5, so b = 1 and j = 1,2,3,4,5.

Multi-logit model built:

$$\ln P_1 = \ln \left(\frac{p(y=1/x)}{p(y=1/x)} \right) = 0 \quad (2)$$

$$\ln P_2 = \ln \left(\frac{p(y=2/x)}{p(y=1/x)} \right) = \beta^2 + \sum_{i=1}^n \beta_i^2 x_i \quad (3)$$

$$\ln P_3 = \ln \left(\frac{p(y=3/x)}{p(y=1/x)} \right) = \beta^3 + \sum_{i=1}^n \beta_i^3 x_i \quad (4)$$

$$\ln P_4 = \ln \left(\frac{p(y=4/x)}{p(y=1/x)} \right) = \beta^4 + \sum_{i=1}^n \beta_i^4 x_i \quad (5)$$

$$\ln P_5 = \ln \left(\frac{p(y=5/x)}{p(y=1/x)} \right) = \beta^5 + \sum_{i=1}^n \beta_i^5 x_i \quad (6)$$

Then the odds of the j -th group relative to the base group can be expressed as:

$$\frac{\pi_{ij}}{\pi_{i1}} = \frac{P(y_i=j/x)}{P(y_i=1/x)} = \exp(x_i' \beta_j) \quad j=2, \dots, 5 \quad (7)$$

The effect of a change in the l -th explanatory variable on this odds ratio can then be expressed as:

$$\frac{\exp(x_i' \beta_j + \Delta x_{il} \beta_{jl})}{\exp(x_i' \beta_j)} = \exp(\Delta x_{il} \beta_{jl}) \quad (8)$$

6 Results

6.1 Factors affecting contribution brackets for rural residents

Tab.9 Results of factors influencing rural residents' choice of pension insurance contribution brackets (multi-logit model) (n=800)

Variable	Contribution behaviour			
	Bracket2/ Bracket1 (Model 2)	Bracket3/ Bracket1 (Model 3)	Bracket4/ Bracket1 (Model 4)	Bracket5/ Bracket1 (Model 5)
Demographic characteristics				
Gender	0.888 (0.171)	1.158 (0.331)	2.394** (0.863)	0.813 (0.328)
Age	1.066*** (0.017)	1.049** (0.023)	1.184*** (0.034)	1.115*** (0.035)
Marital status	0.385** (0.152)	2.340 (1.672)	2.617 (2.226)	4.156 (4.656)
Occupation	2.947*** (0.869)	0.619 (0.219)	2.926** (1.396)	1.271 (0.673)
Edu (CG: Below primary)				
Junior high school	0.613* (0.161)	0.351** (0.157)	0.226*** (0.118)	0.856 (0.500)
High school	1.775* (0.566)	3.106*** (1.345)	2.267 (1.183)	6.535*** (3.858)
College education or above	1.325 (0.565)	2.944* (1.733)	4.588** (2.989)	5.940** (4.330)
Health status	1.445*** (0.194)	0.735* (0.131)	1.041 (0.214)	0.433*** (0.095)
Household characteristics				
Number of children	0.659** (0.113)	0.786 (0.197)	0.441*** (0.136)	0.760 (0.263)
Household size	1.112 (0.104)	1.066 (0.138)	0.963 (0.152)	0.870 (0.169)

Tab.9 (continued)

Log annual per capita household income	1.679** (0.379)	1.239 (0.328)	2.756*** (1.035)	1.169 (0.516)
Total household assets	1.126 (0.095)	0.578*** (0.091)	0.975 (0.151)	1.417** (0.251)
Policy and subjective perceptions (Understanding)				
Content of pension insurance	0.680** (0.110)	0.858 (0.207)	0.964 (0.282)	0.573 (0.199)
Future pension amount	0.744* (0.124)	0.958 (0.240)	0.749 (0.241)	1.266 (0.456)
Policy of paying more and longer will get more	0.942 (0.122)	0.834 (0.167)	0.629** (0.145)	0.956 (0.265)
Destination and operation of funds	1.030 (0.194)	1.148 (0.316)	0.596 (0.204)	1.109 (0.387)
Government subsidies	1.271 (0.209)	0.865 (0.219)	2.095*** (0.580)	1.230 (0.394)
Change procedures	1.457*** (0.206)	1.115 (0.250)	1.320 (0.330)	1.509 (0.457)
Policy and institutional satisfactions (Satisfaction)				
The work of the grassroots government	0.789 (0.135)	1.083 (0.268)	0.563 (0.173)	0.798 (0.272)
The ability to conduct operations	1.557** (0.276)	0.783 (0.188)	1.493 (0.450)	1.495 (0.566)
Effectiveness of publicity	0.941 (0.134)	0.896 (0.198)	0.728 (0.180)	0.956 (0.291)
The setting of contribution brackets	0.999 (0.165)	0.986 (0.235)	0.967 (0.276)	0.768 (0.257)
Incentive policy	0.843 (0.094)	0.790 (0.128)	0.838 (0.151)	1.012 (0.245)
Basic pension payment	1.248* (0.155)	1.635** (0.332)	0.925 (0.216)	1.104 (0.276)
_cons	0*** (0)	0.035 (0.107)	0*** (0)	0* (0.001)

Note: (1) Values in parentheses are standard deviations; (2) *, **, and *** indicate significant at the 10%, 5%, and 1% statistical levels, respectively.

Based on Tab.9, the Demographic characteristics affecting the contribution bracket are analyzed separately and in detail. The gender variable is significant in model 4, with the OR greater than 1. This indicates that men tend to choose a higher contribution bracket than women. The age variable is significant in all models, with the OR greater than 1. As age increases, participants are likely to choose a higher contribution bracket. The marital is significant in model 2, with the OR less than 1. This indicates that widowers tend to choose a higher contribution bracket. The occupation is significant in models 2 and 4, with the OR greater than 1. Other residents are likely to choose a higher contribution bracket than rural residents who work purely in agriculture. The education is significant in all models, but when the education level is Junior high school, the OR is less than 1, while the rest are greater than 1. This indicates that as the education level increases, participants are likely to choose a higher contribution bracket. The reason why the OR is less than 1 is probably that this group is caught in a cognitive dilemma. They mistakenly think that paying for the least expensive insurance is the most economical because they are unaware of the pension insurance policy that applies to urban and rural people. The health has the OR greater than 1 in model 2, but less than 1 in models 3 and 5. This shows that participants are more likely to select a lower bracket as the health status tier increases.

In terms of the demographic characteristics, the OR for the gender is 2.394, which means that male have 139.4% higher odds of choosing contribution bracket 4 than female. The OR for the marriage is 0.385, which means that residents who are not widowed have 61.5% lower odds of choosing bracket 2. The ORs for the occupation are 2.947 and 2.926 respectively, showing that individuals who aren't just farmers are more inclined to select options from brackets 2 and 3. The ORs for junior high school are 0.613, 0.351, and 0.226 respectively, implying that residents with this level of education are more likely to choose bracket 1. The ORs for high school are 1.775, 3.106, and 6.535 respectively, indicating a preference for bracket 5 for this level of education. The ORs for residents with a college degree or more are 2.944, 4.588, and 5.940, respectively, showing that they are more likely to select contribution bracket 5.

The household characteristics that influence the contribution bracket are also analyzed in detail. The number of children is significant in models 2 and 4, with the OR less than 1. As the number of children increases, participants are likely to choose a lower contribution bracket. This phenomenon is related to the concept of

family retirement, which is still favoured by a large proportion of respondents through analysis of the questionnaire (see Fig.7).

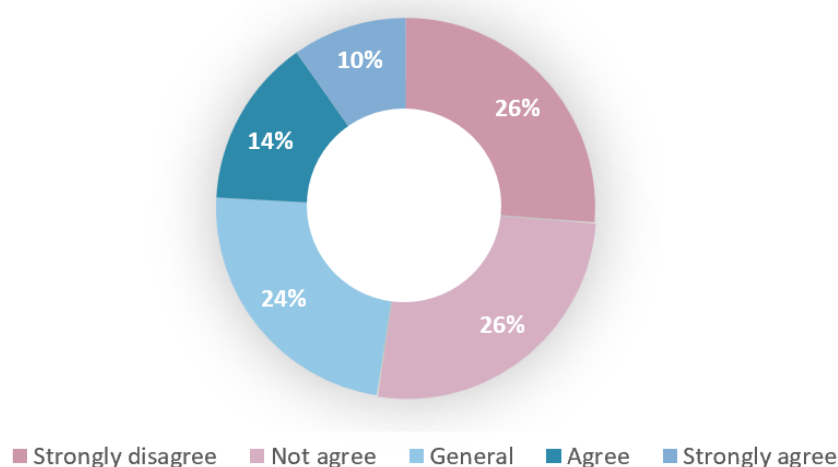


Fig.7 Whether the respondents agree with raising children for old age

The log annual household income per capita is significant in models 2 and 4, with the OR greater than 1. This indicates that participants are likely to choose a higher bracket as their annual household income per capita increases. The total household assets variable has the OR less than 1 in model 3 and an OR greater than 1 in model 5. As total household assets increase further, participants tend to choose a higher bracket.

In terms of the household characteristics, the ORs for the number of children are 0.659 and 0.441 respectively, revealing that the probability of selecting contribution brackets 2 and 4 fall by 34.1% and 55.9%, respectively, with each unit increase in the amount of children. The ORs for the log annual household income per capita are 1.679 and 2.756 respectively, indicating that for each unit increase in annual per capita household income, the odds of residents choosing brackets 2 and 4 increased by 67.9% and 175.6% respectively. The OR for the total household assets is 1.417, which indicates that for each unit increase in total household assets, the odds of residents choosing bracket 5 increased by 41.7%.

From the perspective of policy and subjective understanding, the understanding of pension insurance content is significant in model 2, with the OR less than 1. This suggests that as an understanding of pension insurance content increases, participants are likely to choose a lower contribution bracket. This goes against what could be predicted, probably because participants discover that there isn't a big variation in the amount of pension they receive based on the contribution bracket as they learn more about the pension insurance's contents.

This study briefly calculates the future pension that can be received for each contribution bracket (without calculating the interest on the personal account, 15 years of contributions, using a crediting factor of 139 months to calculate the basic pension that can be received, using the State Council's standard of RMB 70/month).

Tab.10 Future pensions available for each contribution brackets

Payment grade (RMB)	Amount of individual accounts accumulated for 15 years (RMB)	Accumulated subsidy amount for 15 years (RMB)	Monthly pensions available (RMB)	Annual pensions available (RMB)
100	1500	450	84	1008
300	4500	450	106	1267
500	7500	900	130	1565
600	9000	900	141	1695
800	12000	1200	165	1980
1000	15000	1200	187	2239
1500	22500	1200	241	2886

Based on Tab.10 although there is some difference in the annual pensions, there is not a large difference in the monthly pensions received. Therefore, as a rational economic person, one may be inclined to choose the lower contribution bracket simply by considering the rate of return on earnings

Still, regarding policy and subjective perceptions, the understanding of the future pension amount is significant in model 2 with the OR less than 1. The understanding of the policy of paying more and longer will get more is significant in model 4 with the OR less than 1. This implies that as an understanding of the pension amount and this policy increases, there is a tendency to choose a lower bracket. This may be because, despite the different brackets that can be observed in Tab.10, there isn't a significant difference in the amount of pension received. The understanding of government subsidies is significant in Model 4, with the OR greater than 1. As understanding of government subsidies increases, residents are likely to choose a higher bracket. The understanding of the bracket change procedure is significant in Model 2, with an OR greater than 1. As understanding of the bracket change procedure increases, participants are likely to choose a higher contribution bracket.

In terms of policy and subjective perceptions, the ORs for content of pension insurance and future pension amount are 0.680 and 0.744 respectively, indicating that for each increase in understanding level, the odds of residents choosing bracket 2 decrease by 32% and 25.6%. The OR for the policy of paying more and longer will get more is 0.629, indicating that for each increase in understanding level, the odds of residents choosing contribution bracket 4 decrease by 37.1%. The OR for the government subsidies is 2.095, indicating that for each increase in understanding level, the odds of choosing contribution bracket 4 increase by 109.5%. The OR for the change procedures is 1.457, which indicates that for each increase in understanding level, the odds of residents choosing contribution bracket 2 increase by 45.7%.

Regarding satisfaction with policies and institutions, the satisfaction with the ability to conduct business is significant in model 2, with the OR greater than 1. As satisfaction with the level of conduct of business increases, residents are likely to choose a higher contribution bracket. The satisfaction with the basic pension payment is significant in models 2 and 3, with OR greater than 1. This means that as satisfaction with the basic pension payment increases, participants are likely to choose a higher contribution bracket.

In terms of policy and institutional satisfaction, the OR for the ability to conduct operations is 1.557, indicating that for each increase in satisfaction level, the odds of residents choosing contribution bracket 2 increase by 55.7%. The ORs for the basic pension payment are 1.248 and 1.635 respectively, indicating that the odds of choosing contribution brackets 2 and 3 increase by 24.8% and 63.5% respectively for each increase in satisfaction.

6.2 Significance tests and goodness of fit

Tab.11 Significance tests of estimated coefficients

Variable	Wald test	P
Gender	7.890	0.096
Age	47.350	0.000
Marital status	13.680	0.008
Occupation	21.570	0.000
Education	51.270	0.000
Health status	29.550	0.000
Number of children	10.730	0.029

Log annual household income per capita	10.040	0.040
Total household assets	20.270	0.000
Content of pension insurance understanding	7.610	0.107
Future pension amount understanding	4.380	0.357
Policy of paying more and longer will get more understanding	4.460	0.347
Government subsidies understanding	9.030	0.060
Change procedures understanding	8.130	0.087
Satisfaction with the ability to conduct operations	9.940	0.041
Satisfaction with the level of basic pension payment	8.620	0.071

According to Tab.11, it can be found that understanding of the content of pension insurance, future pension amount, and the policy of paying more and longer will get more do not pass the Wald test. The Wald test is met for the following variables: gender, age, marital status, occupation, education, health status, number of children, Log annual household income per capita, total household assets, understanding of government subsidies and change procedures, satisfaction with the ability to conduct operations, and the basic pension payment level.

By assessing the goodness of fit of the model, McFadden's R^2 is 0.195 and the AIC is 2.277. An R^2 between 0.2 and 0.4 is considered as a very good indicator of a good model fit. The simulations by Domenech and McFadden (1975) equate this range to a linear function of 0.7 to 0.9. McFadden's R^2 in this study is close to 0.2 and compares favourably with McFadden's R^2 for similar selections. The smaller value of AIC also demonstrates the goodness of fit of the model.

6.3 Robustness tests

6.3.1 Variable substitution method

The feasibility and scientific validity of the findings of the empirical analysis depend on the stability of the affects of each element on how rural residents behave when selecting their contribution bracket. In the above, the current contribution bracket was chosen as the dependent variable, and this dependent variable is more influenced by the factors in the previous period due to the lagged nature of the effect. In contrast, the contribution bracket that will be selected for the following year reflects the effect of the various factors in the current period. The current contribution brackets is therefore replaced by the next year's contribution bracket.

Tab.12 Results of factors influencing rural residents' choice of pension insurance contribution brackets (multi-logit model) (n=800)

Variable	Contribution behaviour			
	Bracket2/ Bracket1 (Model 2)	Bracket3/ Bracket1 (Model 3)	Bracket4/ Bracket1 (Model 4)	Bracket5/ Bracket1 (Model 5)
Demographic characteristics				
Gender	0.974 (0.185)	0.902 (0.305)	1.778* (0.551)	0.904 (0.244)
Age	1.052*** (0.016)	1.067*** (0.026)	1.097*** (0.025)	1.072*** (0.022)
Marital status	0.396** (0.155)	0.754 (0.561)	1.469 (1.049)	2.453 (1.969)
Occupation	2.111*** (0.570)	1.638 (0.763)	1.810 (0.759)	1.359 (0.489)
Edu (CG: Below primary)				
Junior high school	0.743 (0.193)	1.051 (0.522)	0.383** (0.177)	0.545 (0.202)
High school	1.331 (0.431)	5.639*** (2.848)	2.617** (1.204)	2.410** (0.025)
College education or above	1.018 (0.472)	8.667*** (5.407)	6.942*** (3.843)	3.583** (1.987)
Health status	1.537*** (0.210)	0.710* (0.140)	0.823 (0.146)	0.740* (0.122)
Household characteristics				
Number of children	0.940 (0.160)	0.524** (0.160)	0.605* (0.160)	0.639* (0.151)
Household size	1.351*** (0.124)	1.079 (0.174)	1.155 (0.164)	1.219 (0.154)
Log annual per capita household income	1.846*** (0.398)	0.730 (0.236)	2.367*** (0.790)	1.914** (0.530)
Total household assets	0.908 (0.077)	0.830 (0.131)	0.943 (0.123)	1.789* (0.100)

Tab.12 (continued)

Policy and subjective perceptions (Understanding)				
Content of pension insurance	0.701** (0.111)	0.696 (0.192)	0.720 (0.190)	0.766 (0.181)
Future pension amount	0.721** (0.118)	1.108 (0.315)	1.052 (0.303)	1.112 (0.290)
Policy of paying more and longer will get more	1.061 (0.136)	1.034 (0.236)	0.688* (0.144)	0.524*** (0.105)
Destination and operation of funds	0.992 (0.181)	0.568* (0.183)	0.633 (0.193)	0.751 (0.205)
Government subsidies	1.447** (0.237)	1.559 (0.432)	1.912*** (0.473)	2.175*** (0.492)
Change procedures	1.308* (0.184)	1.637** (0.402)	1.253 (0.283)	1.155 (0.243)
Policy and institutional satisfactions (Satisfaction)				
The work of the grassroots government	0.993 (0.167)	0.702 (0.203)	0.569 (0.156)	0.832 (0.207)
The ability to conduct operations	1.262 (0.218)	1.438 (0.426)	2.337*** (0.649)	1.536* (0.377)
Effectiveness of publicity	1.037 (0.150)	0.817 (0.207)	0.642** (0.139)	0.877 (0.177)
The setting of contribution brackets	0.908 (0.150)	1.210 (0.332)	0.973 (0.244)	0.712 (0.170)
Incentive policy	0.781 (0.085)	0.766 (0.153)	0.797 (0.135)	0.874 (0.138)
Basic pension payment	1.435*** (0.177)	1.425 (0.308)	1.445* (0.300)	1.479** (0.287)
_cons	0*** (0)	0.393 (1.501)	0*** (0v	0*** (0)

Note: (1) Values in parentheses are standard deviations; (2) *, **, and *** indicate significance at the 10%, 5%, and 1% statistical levels, respectively.

Comparing Tab.9 with Tab.12, it was found that household size, the understanding of the destination and operation of funds, and Satisfaction with the effectiveness of publicity also influence residents' contribution behaviour, in addition to the variables mentioned in previously.

Tab.13 Significance tests of estimated coefficients

Variable	Wald test	P
Gender	4.500	0.343
Age	27.880	0.000
Marital status	9.690	0.046
Occupation	8.580	0.073
Education	47.750	0.000
Health status	22.150	0.000
Number of children	9.130	0.058
Household size	11.300	0.023
Log annual household income per capita	16.740	0.002
Total household assets	4.670	0.323
Content of pension insurance understanding	6.680	0.154
Future pension amount understanding	5.200	0.267
Policy of paying more and longer will get more understanding	14.240	0.007
Destination and operation of funds understanding	5.350	0.253
Government subsidies understanding	16.610	0.002
Change procedures understanding	6.520	0.164
Satisfaction with the ability to conduct operations	11.040	0.026
Satisfaction with effectiveness of publicity	5.280	0.260
Satisfaction with the pension payment	12.340	0.015

The bulk of the variables are consistent, as can be shown when comparing the results with Tab.11 and assessing the coefficients for significance. It is vital to highlight that the gender, total household assets, and understanding of the change procedure are no longer significant factors with the addition of the total household size and understanding of the policy of paying more and longer will get more. This may be because a number of factors influenced the contribution bracket behaviour during the year, thus making it more reasonable (see Fig.5).

6.3.2 Change model method

Tab.14 Results of factors influencing rural residents' choice of pension insurance contribution

brackets (multi-probit model) (n=800)

Variable	Contribution behaviour			
	Bracket2/ Bracket1 (Model 2)	Bracket3/ Bracket1 (Model 3)	Bracket4/ Bracket1 (Model 4)	Bracket5/ Bracket1 (Model 5)
Demographic characteristics				
Gender	-0.055 (0.147)	0.090 (0.188)	0.517** (0.217)	-0.052 (0.240)
Age	0.047*** (0.012)	0.030** (0.014)	0.104*** (0.018)	0.065*** (0.018)
Marital status	-0.839*** (0.312)	0.289 (0.459)	0.156 (0.475)	0.683 (0.664)
Occupation	0.613*** (0.211)	-0.405 (0.230)	0.512* (0.288)	0.135 (0.311)
Education	0.146 (0.098)	0.308** (0.120)	0.278** (0.131)	0.469*** (0.139)
Health status	0.267*** (0.100)	-0.188 (0.115)	0.028 (0.129)	-0.469*** (0.135)
Household characteristics				
Number of children	-0.367*** (0.129)	-0.214 (0.167)	-0.563*** (0.188)	-0.197 (0.205)
Household size	0.079 (0.071)	0.039 (0.087)	-0.019 (0.100)	-0.073 (0.114)
Log annual per capita household income	0.417** (0.167)	0.168 (0.182)	0.789*** (0.248)	0.155 (0.259)
Total household assets	0.094 (0.066)	-0.322*** (0.097)	-0.014 (0.098)	0.215** (0.105)
Policy and subjective perceptions (Understanding)				
Content of pension insurance	-0.339*** (0.125)	-0.116 (0.157)	-0.140 (0.178)	-0.348* (0.202)
Future pension amount	-0.211 (0.129)	-0.064 (0.166)	-0.159 (0.194)	0.109 (0.213)

Tab.14 (continued)

The policy of paying more and longer will get more	-0.019 (0.099)	-0.112 (0.126)	-0.189 (0.138)	-0.042 (0.163)
Destination and operation of funds	0.030 (0.144)	0.093 (0.181)	-0.266 (0.213)	0.018 (0.218)
Government subsidies	0.177 (0.126)	-0.119 (0.167)	0.370** (0.177)	0.137 (0.196)
Change procedures	0.284*** (0.108)	0.143 (0.145)	0.207 (0.155)	0.272 (0.178)
Policy and institutional satisfactions (Satisfaction)				
The work of the grassroots government	-0.158 (0.131)	0.057 (0.167)	-0.314 (0.190)	-0.139 (0.203)
The ability to conduct operations	0.284** (0.133)	-0.138 (0.163)	0.253 (0.188)	0.228 (0.218)
Effectiveness of publicity	-0.036 (0.111)	-0.096 (0.144)	-0.158 (0.158)	-0.027 (0.182)
The setting of contribution brackets	0.010 (0.127)	0.036 (0.160)	-0.085 (0.179)	-0.147 (0.200)
Incentive policy	-0.145 (0.087)	-0.206 (0.110)	-0.167 (0.116)	-0.030 (0.143)
Basic pension payment	0.146 (0.096)	0.297** (0.124)	-0.007 (0.140)	0.132 (0.153)
_cons	-7.636*** (1.918)	-2.689 (2.160)	-12.547*** (2.880)	-6.506** (2.990)

Note: (1) Values in parentheses are standard deviations; (2) *, **, and *** indicate significance at the 10%, 5%, and 1% statistical levels, respectively.

Based on the above table, it can be found that in model 4, gender is significant with a positive coefficient. This indicates that men are likely to choose a higher contribution bracket than women. Age is significant in all models, with a positive coefficient. Age and contribution bracket are positively correlated. The marital is significant in model 2 with a negative coefficient, indicating that widowed individuals are likely to choose a higher contribution bracket than those who are not widowed. The occupation is significant in Models 2 and 4, with a positive coefficient. Residents who do not work purely in agriculture are likely to choose a

higher contribution bracket than those who work purely in agriculture. The education, significant in models 3, 4 and 5, has a positive coefficient. Educational and contribution brackets are positively correlated. The health has a positive coefficient in model 2, but a negative coefficient in model 5. This indicates that as the health status tier increases, it is negatively correlated with the contribution bracket.

Additionally, models 2 and 4 with negative coefficients show a significant contribution from the household factor. The contribution bracket is inversely related to the number of children. The log annual household income per capita is significant in models 2 and 4, with a positive coefficient. Annual per capita household income is positively associated with the contribution bracket. The total household assets have a negative coefficient in model 3 and a positive coefficient in model 5. With further increases in total household assets, it is positively correlated with the contribution bracket.

From the perspective of policy and subjective perceptions, the understanding of pension insurance content, which is significant in Model 2, has a negative coefficient. This indicates that understanding of the content of pension insurance is negatively related to the contribution bracket. The understanding of government subsidies, significant in model 4, has a positive coefficient. The understanding of government subsidy is positively related to the contribution bracket. The understanding of the change procedure is significant in model 2 with a positive coefficient. Understanding of the change procedure is positively associated with the contribution bracket.

From the perspective of satisfaction with policies and institutions, satisfaction with the ability to conduct operations is significant in Model 2, with a positive coefficient. Satisfaction with the ability to conduct operations is positively related to the contribution bracket. The satisfaction with the pension payment is significant in model 3. The coefficient is positive. Satisfaction with the pension payment is positively correlated with the contribution bracket.

The results were found to be consistent through the multi-logit model results and the multi-probit results. This implies that there is a robust effect of four factors - demographic characteristics, household characteristics, policy and subjective perceptions, and policy and institutional satisfaction - on the contribution bracket of rural residents.

6.4 Heterogeneity test

When it comes to how the four major determinants affect the rural inhabitants' contribution level, there is intra-group heterogeneity. The concept of raising children for old age is because in the past when China was a small-holder economy for a long time, resources were scarce, so family retirement became the main form of retirement for Chinese residents (Zeng & Lin, 2022). Family retirement has a substitution effect on social retirement (Gao, 2012). Therefore, this thesis uses the classification of whether one agrees with the idea of raising children for old age as a basis for exploring the influence of four main factors on the level of contribution among different groups.

Tab.15 The impact of each factor on whether or not residents agree with the concept of raising children for old age on the contribution bracket (n=800)

Variable	Contribution behaviour							
	Bracket2/Bracket1		Bracket3/Bracket1		Bracket4/Bracket1		Bracket5/Bracket1	
	Agree	Disagree	Agree	Disagree	Agree	Disagree	Agree	Disagree
Demographic characteristics								
Gender	1.131 (0.348)	0.586* (0.173)	1.648 (0.716)	0.678 (0.323)	5.236*** (3.167)	0.797 (0.445)	0.925 (0.673)	0.437 (0.257)
Age	1.055** (0.023)	1.080*** (0.030)	1.017 (0.031)	1.141*** (0.049)	1.225*** (0.054)	1.141** (0.064)	1.108** (0.052)	1.196*** (0.069)
Marital status	0.370* (0.223)	0.627 (0.389)	2.886 (3.488)	1.043 (1.084)	7.621 (9.829)	1.157 (1.092)	1.377 (2.298)	8.676 (2.746)
Occupation	3.021*** (1.285)	3.489*** (1.671)	0.550 (0.286)	1.236 (0.791)	8.197** (7.883)	1.512 (1.058)	1.229 (1.146)	2.703 (2.372)
Education (CG: Below primary school)								
Junior high school	0.380** (0.154)	0.898 (0.363)	0.116*** (0.083)	2.843 (2.084)	0.353 (0.257)	0.093** (0.111)	1.147 (1.165)	0.664 (0.573)
High school	2.319* (1.183)	1.859 (0.919)	1.406 (1.014)	17.662*** (12.756)	3.480 (2.960)	2.299 (2.057)	31.410*** (34.450)	6.480** (5.894)

Tab.15 (continued)

College education or above	1.163 (0.776)	2.216 (1.486)	2.096 (1.847)	12.956*** (12.816)	14.043** (15.506)	4.556 (5.363)	9.941* (12.939)	17.826*** (19.126)
Health status	1.557** (0.309)	1.102 (0.236)	0.905 (0.246)	0.529** (0.154)	1.090 (0.317)	0.929 (0.352)	0.491** (0.175)	0.254*** (0.090)
Household characteristics								
Number of children	0.678 (0.161)	0.591* (0.175)	0.846 (0.295)	0.441 (0.229)	0.479* (0.212)	0.940 (0.515)	0.585 (0.293)	1.190 (0.749)
Log annual per capita household income	1.342 (0.380)	1.253 (0.557)	1.631 (0.652)	0.453 (0.239)	4.255** (2.640)	1.986 (1.415)	0.795 (0.416)	3.140 (2.940)
asset	1.325** (0.166)	1.117 (0.161)	0.535** (0.132)	0.869 (0.229)	0.963 (0.241)	1.112 (0.321)	1.430 (0.437)	2.015** (0.586)
Policy and subjective perceptions (Understanding)								
Government subsidies	1.236 (0.357)	0.938 (0.234)	0.913 (0.413)	0.553 (0.238)	3.007** (1.468)	0.977 (0.459)	0.825 (0.499)	1.211 (0.550)
Change procedures	1.822** (0.482)	1.743*** (0.341)	1.153 (0.481)	1.077 (0.362)	2.107* (0.896)	0.597 (0.225)	1.111 (0.753)	1.717 (0.734)
Policy and institutional satisfaction (Satisfaction)								
The ability to conduct operations	1.602 (0.522)	1.253 (0.309)	0.411** (0.175)	1.180 (0.508)	0.802 (0.473)	1.341 (0.673)	2.062 (1.691)	1.081 (0.571)

Tab.15 (continued)

Basic pension payment	1.238 (0.305)	1.082 (0.186)	1.458 (0.635)	1.522 (0.408)	0.740 (0.343)	1.301 (0.425)	1.053 (0.572)	1.235 (0.410)
_cons	0** (0.001)	0* (0.001)	0.006 (0.026)	2.606 (16.157)	0 (0)	0 (0.001)	0.012 (0.086)	0 (0)

Note: (1) Values in parentheses are standard deviations; (2) *, **, and *** indicate significance at the 10%, 5%, and 1% statistical levels, respectively.

According to the Tab.15, four main factors influence the contribution brackets of rural residents with different views on whether they agree with the idea of raising children for old age. Male residents who do not agree with it tend to choose a lower contribution bracket when choosing between Contribution Bracket 1 and Contribution Bracket 2. As a result of their greater openness, the males in this group prone to select a higher level for their spouses.

There is a slight difference in the effect of age on contribution brackets between those who agree or disagree with the concept of raising children for old age. Residents who disagree with it are more willing to pay a higher contribution bracket as they get older than those who agree with the idea. This is because elder locals who are opposed to the proposal are more likely to rely on this old-age insurance to support themselves in old life.

There is also a slight difference in the impact of work on contribution brackets for residents who agree or disagree with the idea of raising children for old age. The impact of work is higher for residents who disagree with the idea than for those who agree with the idea. This is because disapproval of child-raising is more prevalent among those engaged in non-agricultural activities, thus further reinforcing the impact of work.

There is a difference in the impact of health on the contribution bracket between residents who agree or disagree with the concept of raising children for old age. In choosing contribution bracket 5 versus 1, the impact of health on the contribution bracket is less for residents who agree with the idea than for those who do not agree with the idea. This is because residents who do not agree with the idea are more likely to prefer social old age to family old age. As a result, when their health deteriorates, they choose to pay a higher contribution bracket to secure their old age.

There is a difference in the impact of total household assets on contribution brackets for residents who agree or disagree with the concept of raising children for old age. In the choice of contribution bracket, 5 versus 1, residents who do not agree with the idea are more likely to choose a higher contribution bracket as their assets increase. This is because this group tends to choose social retirement, and when assets rise further, they are willing to invest more in it.

There is also a slight difference in the impact of the understanding of the change process on the contribution bracket between those who agree with the concept of raising children for old age and those who disagree with the idea. When choosing contribution brackets 2 or 1, residents who agree with the idea are more strongly influenced by their understanding of the change process. This is caused by the fact that inhabitants who support the notion are more inclined to alter their contribution choices, making them more intimately related to the comprehension of the change process.

It is crucial to highlight that when people who oppose the idea of having children for old age choose their contribution bracket, factors such as marital status, per capita household income, understanding of government subsidies, and satisfaction with the basic pension are no longer taken into account. This may be because this group, which prefers social pensions, so being widowed or not, finances, understanding of financial subsidies, and satisfaction with the basic pension payment do not really change their perception of old age.

7 Suggestions

By combing through the literature and combining the results of data analysis and interviews obtained from field research, this thesis proposes corresponding suggestions for solving the dilemma of low contribution brackets for rural residents in Tengzhou.

7.1 Suggestions based on the individual participant

With the promotion of family planning, although this policy has now been changed, the development of smaller family size has been the rule. In addition, the traditional Chinese idea of "raising children for old age" is deeply rooted, so that one child has to bear the burden of two parents in old age. In addition, China's urbanization process is accelerating, and a large number of young and middle-aged people are moving into the cities, which has led to a sharp increase in the number of empty nesters, and the traditional family retirement system has been affected. Under this circumstance, should change the participants' concept of family pension through propaganda and education to increase the enthusiasm of participating in social pension insurance. It is also required to boost the level of contribution, encourage early enrolment in social old-age insurance, and raise participant knowledge of concerns about their old age.

According to the results of field research, the vast majority of inhabitants have a vague understanding of this old-age insurance. Because they tend to follow trends rather than starting with their own pension needs, the majority of them opt for lower contribution levels. Therefore, it is necessary to increase publicity, not only through new media, but also through grassroots staff. Care must be made to ensure that the propaganda process uses straightforward language because rural populations typically tend to be older and less educated than urban inhabitants. In order for people living in rural areas to properly understand crucial concepts like the number of years of contributions, the contribution tiers, the pension payment methods, the pension amounts, and government subsidies. Rural inhabitants should have more access to information on this pension insurance in the meantime, and a special hotline and Q&A department must be set up to answer their concerns.

7.2 Suggestions based on grassroots government

The local level of government assists in the establishment of this pension insurance. In addition to being in charge of the specific insurance issues, they are also required to notify the residents about the pertinent policies. Therefore, this thesis also makes relevant suggestions from the grassroots government level. Through econometric analysis, it is found that residents' education, personal income, and total household assets are closely related to the choice of contribution bracket. As a result, local government should prioritize education in country places and address the disparity in educational resources between cities and rural regions. On the basis of accelerating the construction of modern education in rural areas and improving the quality of nine-year compulsory education in rural areas, the government should strengthen the

investment in rural high school education to improve the education level of rural residents. For young and middle-aged residents, adult re-education training and vocational education can also be developed, thus improving the skill level and knowledge of rural residents. The degree of education is also closely associated to personal income and total household assets, so raising the level of education can also help to raise personal income and total family assets. It is also necessary to take multiple ways to help develop the rural economy, which can increase financial support to rural areas, improve rural infrastructure construction and help rural residents to invest in factories in order to improve their ability to pay contributions.

In the results of the data analysis, policy perception and personal subjective perception as well as policy and institutional satisfaction also influence the choice of contribution grade. When citizens pay for urban and rural inhabitants' pension old-age insurance, the grassroots government can provide residents with pamphlets outlining the insurance transfer and other crucial information to help them better understand the policy. This is done in conjunction with the theory of knowledge, attitude, and practice. The grassroots government can further its policy propaganda in other ways, such as by sending staff members into rural areas to allay residents' concerns and convince them that the pension insurance systems will provide them with better long-term care. The grassroots government also needs to promote rural residents' happiness and trust in its work, strengthen the development of a service-oriented government, and build a positive government image. To strengthen the training of the grassroots staff and regularly assess the staff's business handling ability and attitude, so as to change the impression of arrogance and rudeness of the government staff in the minds of rural residents in the past year. Through the change of "knowledge" and "attitude", the "practice" will be changed, thus changing the residents' behaviour of choosing the payment grade.

Since China is a "human society", residents are greatly influenced by the people around them in the process of choosing their contribution level. The survey found that family members, friends and relatives, and village cadres have a deeper influence on the residents, so the grassroots government should make reasonable use of the "human society" and play a positive herd effect. The village cadres should receive training so they may better understand this old-age insurance and promote the notion that the higher the contribution level, the better the protection of their old age. In order to influence those around them positively, it is also feasible to simultaneously increase the villagers' knowledge of pension insurance through propaganda to the more illustrious villagers and inspire them to choose the contribution level best suited to their own pension needs based on their own realities.

7.3 Suggestions based on system

Based on the public product theory, the quasi-public product of this old-age insurance has beneficial externalities. However, the operational effect of the system does not meet expectations, and further improvement is needed in the system design and incentive mechanism. The study reveals that the choice of contribution grade is positively impacted by the rise in satisfaction with the amount of basic pension payment. Therefore, the subsidy standard should be

reasonably increased so that the pension can meet the basic pension needs of rural residents. However, the problem cannot be completely solved by raising the subsidy standard alone, and further optimization of the contribution bracket setting is needed. With the improvement of rural residents' income, except for some special groups, they can fully pay the minimum contribution level of 300 RMB per year, so the minimum contribution level can be moderately increased in the design of the contribution level. At present, Tengzhou City sets twelve contribution brackets, which can be moderately relaxed in the future. Currently, several provinces employ dynamic contribution brackets, which are constantly updated in accordance with the annual per capita income and economic development of the region. This has given Tengzhou city some experience to draw from.

Raising the degree of coordination is important because the various economic scales in Tengzhou City's regions cause a gap in the ability to offer public goods. A higher level of coordination can better coordinate the management of pension insurance funds and eliminate the differences between regions. Therefore, on the basis of the current county-level coordination, it is possible to gradually raise the level of coordination, transitioning from the county level to the municipal level, then to the provincial level, and finally to achieve nationwide coordination and management. At the same time, raising the level of coordination can also better supervise the use of pension insurance funds. Under the premise of safeguarding the safety of pension insurance funds, the investment portfolio will be used to obtain higher returns. The gains will then be directly reflected in individual accounts, allowing residents to more intuitively feel the increase in their account funds, thus promoting them to choose higher contribution levels.

The fairness theory states that in order to safeguard low-income people, the urban and rural inhabitants' old-age insurance system should emphasize justice and differentiate. The current incentive mechanism is not perfect, and the subsidies for each contribution bracket are different but the difference is small. This will not activate the motivation to contribute and should be differentiated for different contribution brackets. If the accumulated contributions to the pension insurance exceed 15 years, the basic pension will be increased by 2 yuan per month for each additional year of contributions. This means that an additional year of contribution will only increase the basic pension by 24 yuan. Therefore, government subsidies and basic pensions, while focusing on fairness, should also focus on the fairness of high contribution residents and not just pursue absolute equality. As a result, the system should be designed with an emphasis on reducing the difference between the high contribution bracket and the low contribution bracket's investment rate of return and the number of years it takes to recover expenditures.

8 Discussion

The choice of contribution bracket is influenced by four factors: demographic characteristics, household characteristics, policy and subjective perceptions, and policy and institutional satisfaction. This was discovered through field research on the current state of old-age insurance for urban and rural inhabitants in Tengzhou City using the multi-logit model.

Contrary to the findings of this thesis, Lin (2017), based on research in the Henan Province, utilized a logistic regression to conclude that gender, age, and health had no discernible influence on the bracket. The study's findings indicate that men tend to select higher brackets than women do, that brackets are favorably correlated with age and adversely correlated with health. The reason for this difference is most likely due to the difference in sample and model selection. Lin (2017) only divided the sample into those who chose the lowest contribution bracket and those who did not choose it, with a weighting of 76.4% and 23.6% respectively. So this sample could not capture the effect of different factors on brackets. Tengzhou is a fourth-tier city with weaker economic and cultural growth and a stronger tradition of favoring men over women, which can also be demonstrated in the local gender ratio, which accounts for the stronger willingness of men to participate in this thesis (see Tab.2). As residents get older, their need for old age is stronger, and also as their health decreases, their sense of worry increases, so they will tend to choose a higher bracket.

Wen et al. (2014), based on research in Shanxi Province from 2011-2012, used multiple regression and ANOVA to find that widowed households were more likely to choose a lower contribution bracket due to financial constraints. However, this study contradicted this study and found that widowed families were likely to choose the 2nd brackets compared to the lowest bracket. This is because Shandong Province is the home of Confucius and Mencius and is more influenced by Confucian culture than Shanxi Province, and is also more inclined to live in stability and have a strong sense of worry. At the same time, although widowed families are less well-off, there is less difference in the amount paid between bracket 1 and bracket 2, and they are less constrained by financial constraints.

Cheng (2014) used an ordered probit model based on a sample in Chengdu and found educational was positively related to bracket. Residents who were not engaged in agricultural production activities were more willing to choose a higher contribution bracket. This supports earlier discoveries. It is significant to mention that higher education levels are associated with better pension understanding and frequently more financial resources. Those who are solely engaged in agricultural activities tend to be at a financial disadvantage.

Wang and Guo (2016) discovered that one-child households and households with greater annual per capita household income were willing to choose a higher contribution band. They did this using a binary logistic regression model relying on an empirical investigation in ten cities in Shanxi province. The outcomes of this investigation are supported by these findings. This is because, on the one hand, having more children increases the financial strain on the family, which makes parents more motivated to put money away. However, a sizable portion of respondents in Tengzhou continue to favor family pensions (see Fig.7). The substitution effect of family pensions for social pensions grows as the number of

children rises, preventing residents from selecting a higher contribution bracket. Furthermore, the household's economic strength is strongly correlated with the annual per capita income, and the better the economy, the more free money is available to pay a higher contribution bracket. And this study also introduces total household assets into the household factor, which is positively related to the level of contribution. This is because although total household assets are related to income, they are somewhat different. Total household assets better reflect the overall economic strength of a household.

Xu and Xue (2016) used an ordered logistic regression model based on survey data from 464 farmers in Junan County and found that residents were willing to choose a higher contribution bracket when their understanding of pension insurance reached general level. It is crucial to notice, though, that the factors in their analysis were too generic when comparing the studies. When asked this question, it is challenging for respondents to provide precise responses, which has a significant impact on the model's accuracy. There are multiple aspects of perceptions of basic pension insurance, so this study divides perceptions into six major aspects to more accurately describe the perceptions. Finally, it was found that the understanding of government subsidies and change procedure were positively related to the contribution bracket. This is because the more knowledgeable one is about government subsidies, the better one understands the advantages of the insurance. The more you know about the process of changing contribution brackets, the more you will be able to eliminate worries about transferring insurance and choose a more appropriate contribution bracket.

Du (2021), using a logistic regression model based on household study results from a prefecture-level city in Henan province, discovered that the more satisfied people were with the service provider and the policy design, the higher the chosen contribution band. Similar results were obtained in this study. In order to better analyse the impact of policy and institutional satisfaction on contribution brackets, six perspectives were set up for this study, and finally it was found that satisfaction with the ability to conduct operations and pension payment were positively related to contribution brackets.

Earlier studies have mostly divided the group into whether choose the lowest bracket. This does not adequately account for the effects of having different contribution brackets. This study therefore uses a multi-logit model to classify contribution brackets into five categories, which makes the study more reasonable. At the same time, most previous studies have been based on the analysis of individual and household characteristics, ignoring the influence of personal knowledge and attitude. Therefore, the focus of this study is on how contribution brackets are affected by policy and subjective perception, policy and institutional satisfaction.

In addition, there are certain limitations to this study. The data is individual data of participants and fail to integrate with macro policies. There is a lack of integration with macro-level policy data on ageing, finance, health, etc. Therefore, in the following research, this study can be used as a basis and relevant policy instrument variables can be added. Through empirical analysis, the effectiveness of the macro policy design can be further tested. The ordered logit option better fits the dependent variable in this study, whereas the multi-logit model loses some effectiveness by ignoring the relative merits of each classification. In the next study, a broader sample of the survey is obtained and optimised.

9 Conclusion

In view of China's developing significant ageing problems, improving this old-age insurance system is useful in addressing the issue of rural households' pensions and in encouraging social fairness. This study determines that there is a contribution dilemma in Tengzhou City based on field survey data from 2019 and utilizes a multi-logit model to analyze the variables influencing rural residents' choice of insurance payment levels. Accordingly, the thesis draws the following research conclusions.

Rural inhabitants' choice of urban and rural residential insurance contribution bracket is impacted by demographic factors, namely by gender, age, marital status, level of education, and health. Men and widows are likely to pay higher bracket. The contribution bracket is positively correlated with age and education, and inversely correlated with health.

The number of children, annual per capita household income, and total household assets are the key determinants of household characteristics that affect rural residents' choice of urban and rural residential insurance contribution bracket. The contribution bracket is favorably correlated with both yearly per capita household income and total household assets, and negatively correlated with the number of children.

The decision of which contribution bracket to pay for rural citizen is influenced by policy and subjective perceptions. The understanding of pension subsidies and the understanding of the changing procedure are positively related to the contribution bracket.

Satisfaction with policies and institutions has an impact on contribution brackets. Satisfaction with the ability to conduct operations and satisfaction with pension payment are positively related to the contribution bracket.

External factors also influence the contribution bracket. Rural residents are influenced by family members, relatives, friends and village cadres, and the new rural cooperative health insurance. The influence of family members is the strongest, and the influence of the new rural cooperative medical insurance is the weakest.

Based on the above findings, this thesis applies Public Product Theory, Fairness Theory and Theory of Knowledge, Attitude and Practice to put forward corresponding suggestions at the individual level, grassroots government and system level. To solve Tengzhou's contribution problem, ensure this old-age insurance system's long-term sustainability, and transform traditional family pensions into modern social pensions that are more logical and scientific in character.

It should be emphasised that this study explores the influencing factors that affect the choice of contribution bracket by rural residents in Tengzhou. Urban residents in Tengzhou are therefore excluded from the sample. The recommendations presented in this thesis are solely applicable for Tengzhou because each territory has a different social and cultural background, as well as a varied degree of economic development. Although there are some implications for other cities, they still need to be adapted to local conditions.

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Appendix

Questionnaire

Interviewee scope: rural residents aged 20 (excluding students) to 59 who are eligible to participate in resident endowment insurance in their domicile place

● Basic personal information and family information

Q1. Your gender :

1. Male 2. Female

Q2. Age : __aged

Q3. Your level of education is :

1. Illiterate or very few 2. Primary school 3. Junior high school

4. High school/ Junior college 5. Junior college and above

Q4. Your Marital Status :

1. Unmarried 2. Married 3. Divorced 4. Widowed

Q5. Your career:

1. Farmer 2. Worker 3. Part work, Part farming 4. Self-employed

5. Unemployed 6. The other

Q6. Your health:

1. Very poor 2. Less 3. General 4. Better 5. Very good

Q7. Are you joining the urban and rural residents endowment insurance now :

1. Yes 2. No

Q8. If you have already participated, your payment level is :

1. 100 yuan 2. 300 yuan 3. 500 yuan 4. 600 yuan 5. 800 yuan

6. 1000 yuan 7. 1500 yuan 8. 2000 Yuan 9. 2500 Yuan

10. 3000 Yuan 11. 4000 Yuan 12. 5000 Yuan

Q9. Are you willing to participate in the basic endowment insurance for urban and rural residents next year?

1. Yes 2. No

Q10. If you choose to participate in the basic endowment insurance for urban and rural residents next year, the payment level is: __RMB (Q8 levels for reference)

Q11. In 2019, the contribution amount for your personal participation in the new rural cooperative medical insurance is: __RMB

Q12. Would you like to participate in the new rural cooperative medical care system next year?

1. Yes 2. No

Q13. If you like, you will pay: __RMB next year

Q14. Which payment level would you like to choose next year for the following new rural cooperative medical care system :

1. 180 Yuan 2. 220 Yuan 3. 260 Yuan 4. 00 Yuan 5. 340 Yuan

6. 380 Yuan 7. 420 Yuan 8. 460 Yuan 9. 500 Yuan

Q15. How many members of your family will participate in the endowment insurance for urban and rural residents in 2019? __people , paid a total of __yuan

Q16. How many members of your family will participate in the new rural cooperative medical insurance in 2019? __persons , paid a total of __yuan

Q17. The total number of your family is: __people.

(Among them, __people over the age of 60 and __under the age of 16)

Q18. The number of your children is: __ people (among them, __boys, __girls).

Q19. Whether you or your family members were hospitalized last year :

1. Yes 2. No

Q20. If you are hospitalized, total cost: __ RMB, reimbursement: __RMB

Q21. If you or your family members visit the outpatient department for __ times in the last year, the total cost is __ RMB, and the reimbursement expense is __ RMB.

Q22. Do you or any of your family members suffer from chronic diseases :

1. Yes 2. No

Q23. Whether you or your family members purchase commercial endowment insurance or medical insurance :

1. Yes 2. No

Q24. The number of acres of land your family owns is: __mu.

Q25. Your family loan amount: __RMB (fill in 0 without loan)

Q26. Your personal income last year is: __yuan. Your family's income last year was: __ yuan, and your expenditure was: __ yuan.

Q27. Your family's main source of income :

1. Breeding and planting 2. Migrant work 3. Individual operating income
4. Government aid 5. Investment income 6. Other

Q28. The value of your family housing is about :

1. 0-100000yuan 2. 100000-200000yuan 3. 200000-300000yuan
4. 300000-400000yuan 5. 400000 yuan or more

Q29. Does your family have a car?

1. Yes 2. No

Q30. Does your family have a color TV?

1. Yes 2. No

Q31. Does your family have a washing machine :

1. Yes 2. No

Q32. Does your family have a refrigerator :

1. Yes 2. No

Q33. Your family total assets about :

1. 0-200000yuan 2. 200000-400000yuan 3. 400000-600000yuan
4. 600000-800 000yuan 5. 800000yuan more

● Policy cognition and personal subjective cognition

Q34. How do you understand the endowment insurance policy for urban and rural residents :

1. Television broadcasting 2. Newspaper 3. Mobile 4. Village cadres
5. Relatives and neighbours

Q35. Do you understand the specific content of the endowment insurance for urban and rural residents :

1. Very poorly understood 2. Not understand 3. General
4. Understand 5. Very understand

Q36. Do you know how much pension you will be able to receive in the future:

1. Very poorly understood 2. Not understand 3. General
4. Understand 5. Very understand

Q37. Do you know that with the increase of individual payment level and payment period of the endowment insurance for urban and rural residents, the pension will increase accordingly?

1. Very poorly understood 2. Not understand 3. General
4. Understand 5. Very understand

Q38. Do you know where your pension money for the endowment insurance for urban and rural residents is going or how it works?

1. Very poorly understood 2. Not understand 3. General
4. Understand 5. Very understand

Q39. Do you know the amount of government subsidy corresponding to different payment levels :

1. Very poorly understood 2. Not understand 3. General
4. Understand 5. Very understand

Q40. Do you know the procedure of change of payment level

1. Very poorly understood 2. Not understand 3. General
4. Understand 5. Very understand

Q41. If the government increases the subsidy or increases the basic pension, will you increase the contribution level :

1. Yes 2.No

Q42. Are you satisfied with the current level of basic pension provided by the old-age insurance for urban and rural residents :

1. Very dissatisfied 2. Not satisfied 3. General 4. Satisfied
5. Very satisfied

Q43. Is the payment level of endowment insurance for urban and rural residents that you currently choose already your economic limit?

1. It can be higher 2. It is affordable 3. It has reached its limits
4. It has not been measured

Q44. Do you think the current pension insurance benefits for urban and rural residents can meet your basic life in old age :

1. Absolutely not 2. Can't 3. General 4. Can 5. Absolutely can

Q45. How much pension do you think you need to get every month to meet your expectation :

1. Less than 500 yuan 2. 501-1000 yuan 3. 1001-1500 yuan
4. 1501-2000 yuan 5. More than 2000 yuan

Q46. Do you think it is cost-effective for you to choose a higher level endowment insurance for urban and rural residents :

1. Very not worthwhile 2. Not worthwhile 3. General
4. Economical 5. Very economical

Q47. Do you agree with the idea of "raising children for old age" :

1. Very disagree 2. Not agree 3. General 4. Agree 5. Very agree

Q48. Do you agree with the concept of "land for the aged"?()

1. Very disagree 2. Not agree 3. General 4. Agree 5. Very agree

● The influence of external factors

Q49. What are your main reasons for choosing the current payment level :

1. Propaganda of village cadres 2. Choice of relatives and friends
3. Personal economic strength 4. Degree of cost-effective policy

Q50. Will you be affected by the family, when choose to participate in the urban and rural residents endowment insurance level ?

1. Not at all 2. A little 3. Very much 4. Completely

Q51. Do you choose to participate in the urban and rural residents endowment insurance level will be affected by relatives and friends :

1. Not at all 2. A little 3. Very much 4. Completely

Q52. Do you choose to participate in the urban and rural residents of the pension insurance level will be affected by the village cadres publicity and encouragement :

1. Not at all 2. A little 3. Very much 4. Completely

Q53. Will your choice of payment level for urban and rural residents' endowment insurance be affected by the payment of the new rural cooperative medical insurance?

1. Not at all 2. A little 3. Very much 4. Completely

Q54. Your friends and relatives around you tend to what kind of pension:

1. Land pension 2. Raising children for old age 3. Self pension
4. Social pension

● **Satisfaction with policies and institutions**

Q55. Are you confident that the pension insurance for urban and rural residents can be paid on time and in full in the future?

1. Very poor confident 2. No confident 3. General 4. Confident
5. Very confident

Q56. Are you satisfied with the implementation of basic endowment insurance for urban and rural residents by grassroots governments :

1. Very dissatisfied 2. Not satisfied 3. General 4. Satisfied
5. Very satisfied

Q57. Are you satisfied with the level of service handled by the staff of the agency:

1. Very dissatisfied 2. Not satisfied 3. General 4. Satisfied
5. Very satisfied

Q58. Are you satisfied with the publicity effect of local basic government on endowment insurance for urban and rural residents :

1. Very dissatisfied 2. Not satisfied 3. General 4. Satisfied
5. Very satisfied

Q59. Are you satisfied with the current payment levels of endowment insurance for urban and rural residents :

1. Very dissatisfied 2. Not satisfied 3. General 4. Satisfied
5. Very satisfied

Q60. Are you satisfied with the current incentive policy of "pay more get more" and "pay more long time get more" of the endowment insurance for urban and rural residents :

1. Very dissatisfied 2. Not satisfied 3. General 4. Satisfied
5. Very satisfied